Physics-constrained generative models of a parametric MHD dynamo

M2 internship report · race pilot, team claude · a comparative study of constraints imposed by design, loss and sampling

Table of contents

Abstract **4**

1 Introduction **5**
1.1 Context: the internship offer and the MilaDy project . . . 5
1.2 Question and approach . . . 5
1.3 Contributions . . . 7
1.4 Outline . . . 7

2 State of the art **8**
2.1 Diffusion and flow matching . . . 8
2.2 Generative models of flows and of PDE solutions . . . 8
2.3 Three places for a physical constraint . . . 8
2.4 Helical dynamos in periodic boxes . . . 9
2.5 Metrics . . . 9
2.6 What was not found . . . 9

3 Methods **10**
3.1 The physical system . . . 10
3.2 Stationarity identities . . . 10
3.3 Latent representation and generative model . . . 11
3.4 Constraint mechanisms . . . 12
3.5 The linear part of the velocity field . . . 15
3.6 Metrics . . . 15

4 Experiments **18**
4.1 Reference data . . . 18
4.2 Protocol . . . 20
4.3 Compute . . . 22

5 Results **25**
5.1 The registered protocol: every first-generation model is far from the reference . . . 25
5.2 Diagnosis: the models do worse than a Gaussian field . . . 26
5.3 Giving the field its linear part . . . 28
5.4 Two Gaussian laws: what the covariance between modes holds . . . 30
5.5 Beyond a Gaussian law: three tests . . . 33
5.6 Loss and sampling mechanisms on a model close to the reference . . . 37
5.7 Longer trainings and wider networks . . . 38
5.8 Topology: null points and the small scales . . . 43
5.9 Replication with other data . . . 43
5.10 Objectives of the race . . . 45

6 Comparative study **47**

7 Toolbox **55**

8 Discussion **58**
8.1 What each place of constraint buys 58
8.2 Negative results and refuted claims 59
8.3 Limits 60
8.4 What the incidents of the pilot teach 61

9 Conclusion **62**

10 References **64**

11 Reproducibility **66**
11.1 Commands 66
11.2 Result files and commits 66
11.3 GPU jobs and costs 68
11.4 Figures of this report 69

Status of this document This report is the deliverable of team claude in the race pilot of this repository (`race/DELIVERABLES.md`). It is written in the form of the M2 internship report that the MilaDy offer asks for, by an AI agent working alone for one session, and it reports one experiment, E4, designed, run and reviewed during that session. Nothing in it has been read or validated by the supervisors of the internship, and no data of the MilaDy consortium was used.

Three statuses are kept apart throughout. What is **measured** comes from result files written by committed scripts of this repository (`experiments/results/`, summarised in `experiments/results/digest.md`); every table of numbers below is generated from those files by `experiments/e4_report_tables.py`. What is **published** is cited with a reference taken from the verified source files `docs/specs/sources/`. What is **conjectured** is said to be so. The system studied is small (a 16^3 grid, 665 Fourier modes, Reynolds number 10): the conclusions are about the mechanisms, not about turbulent dynamos.

The HTML version of this report and an interactive explorer of the results are published with the site of the repository, under `teams/claude/report/` and `teams/claude/explorer/`.

Abstract

The MilaDy internship offer asks for generative models of parameter-dependent 3D magneto-hydrodynamic fields in which physical constraints are imposed through model design, training losses or sampling procedures, and for a comparative study of these strategies. This report carries out that comparison on a system small enough to be simulated and learned many times: a helically forced dynamo in a periodic box (16^3 grid, 665 Fourier modes, $Re = 10$), conditioned on the magnetic diffusivity η, with training, validation and held-out values of η, five model seeds per variant, and hypotheses registered with a refutation criterion before each computation. Conditional flow-matching models on Fourier modes, with constraints placed by design, by the loss and at sampling time, at several widths and training lengths, are compared with a diffusion model and two Gaussian baselines on an energy distance to the reference law relative to its noise floor, the divergence, the total and magnetic energy balances, spectra and helicity, the number of magnetic null points, a restart test in the solver and a polarity test.

All results are **measured** in this repository. Zero divergence is exact by design (vector potentials) or by projection, while a divergence penalty fails to enforce it. These mechanisms do not decide the quality of the law: every first-generation model is more than a hundred noise floors from the reference, and the plain networks fit the data worse than a trivial Gaussian velocity field. Giving the velocity field an explicit mode-wise affine term reduces the distance by a factor of about twenty, to nine floors. Two Gaussian laws without a network are closer still in energy distance. A Gaussian law per Fourier mode is at 3.5 floors but has no magnetic production; a Gaussian law that keeps the covariance between modes, added after an external review, is at 2.5 floors, with a magnetic energy balance residual of 0.07 and half the null-count error of the first. Three tests registered after the review of the third version measure what the best learned model holds beyond a Gaussian law. It carries a third-order part of the magnetic production, about half of that of the reference at two of the three held-out values and a small part of the opposite sign at the third; the energy distance of the objectives barely separates the model from the Gaussian law fitted to its own samples. Restarted in the solver, the laws tested that start on the reference magnetic energy or below it all gain 10 to 21 % of it: the learned model, which starts 12 % low, comes back to the reference, and the Gaussian laws fitted to the data, which start on it, overshoot it by 9 and 17 % before coming back towards it; the restart drift of the objectives, relative to each law's own energy, confounds the two and does not rank the laws. A law-level stationarity loss lowers the residual of the energy balance at parameter values without data but, on a model close to the reference, degrades the law; exponential tilting on the same identities at sampling time improves that model on most metrics. No generator close to the reference reproduces the magnetic topology, and none produces states that the solver accepts as stationary. Twenty-four hypotheses were registered and decided for this experiment, each with a refutation criterion written before its computation; the refuted ones are reported as such. The code (fifteen tested constraint mechanisms, two of them not yet used in a full run, a diffusion sampler, a null-point finder) is in the project toolbox.

1 Introduction

1.1 Context: the internship offer and the MilaDy project

The internship offer that this report answers is entitled *Physics-Constrained Generative AI Modelling of Parametric Dynamical Systems*. It belongs to the ANR project MilaDy, on machine-learning models of turbulent dynamos, and asks for generative models (diffusion or flow matching) of parameter-dependent three-dimensional fields in which physical constraints are imposed "through model design, training losses or sampling procedures". The models are to be evaluated with energy spectra, helicity, divergence control and the preservation of topology, and the expected outcome is a comparative study of the strategies together with code integrated into the project toolbox (`docs/specs/sources/supervisors.md`, §1.2).

A dynamo is a flow of an electrically conducting fluid that sustains its own magnetic field. The equations are those of magnetohydrodynamics (MHD): the Navier–Stokes equations with the Lorentz force, coupled to the induction equation. Two control parameters matter here, the kinetic Reynolds number Re and the magnetic Reynolds number Rm, which measures induction against magnetic diffusion. Simulating a dynamo at one parameter value is expensive, and the scientific questions (thresholds, saturation levels, statistics of the saturated state) are questions about how the statistically stationary state changes with the parameters. A conditional generative model addresses exactly that object: it learns the law of the snapshots of the stationary state as a function of the parameter and can then draw new snapshots at parameter values that were never simulated.

The difficulty, and the subject of the offer, is that a generated field must be physically admissible. The magnetic field must be divergence-free. The energy injected by the forcing must, on average, be dissipated. The symmetries of the equations must be those of the generated law. The topology of the magnetic field, for instance its null points, should be that of the reference. The published literature offers three places where such requirements can enter a generative model, which are the three places named by the offer: the design of the model, the training loss, and the sampling procedure.

1.2 Question and approach

The question of this report is narrow and practical: **for a parametric 3D MHD dynamo, which constraints should be imposed where, and what does each mechanism buy on the metrics the offer names?** The approach is a controlled comparison on a system small enough to be simulated, learned and evaluated many times: a helically forced dynamo in a periodic box, on a 16^3 grid, with the magnetic diffusivity η as the conditioning parameter (Figure 1.1). On this system the reference law can be sampled at will, every quantity can be computed to round-off, and each comparison can be repeated over five model seeds.

The work follows the protocol of the repository (`AGENTS.md`): each hypothesis is written in a register with a quantitative refutation criterion before the computation; each quoted comparison uses five seeds; a reviewing role, the `encadrant`, recomputes the result and gives the verdict;

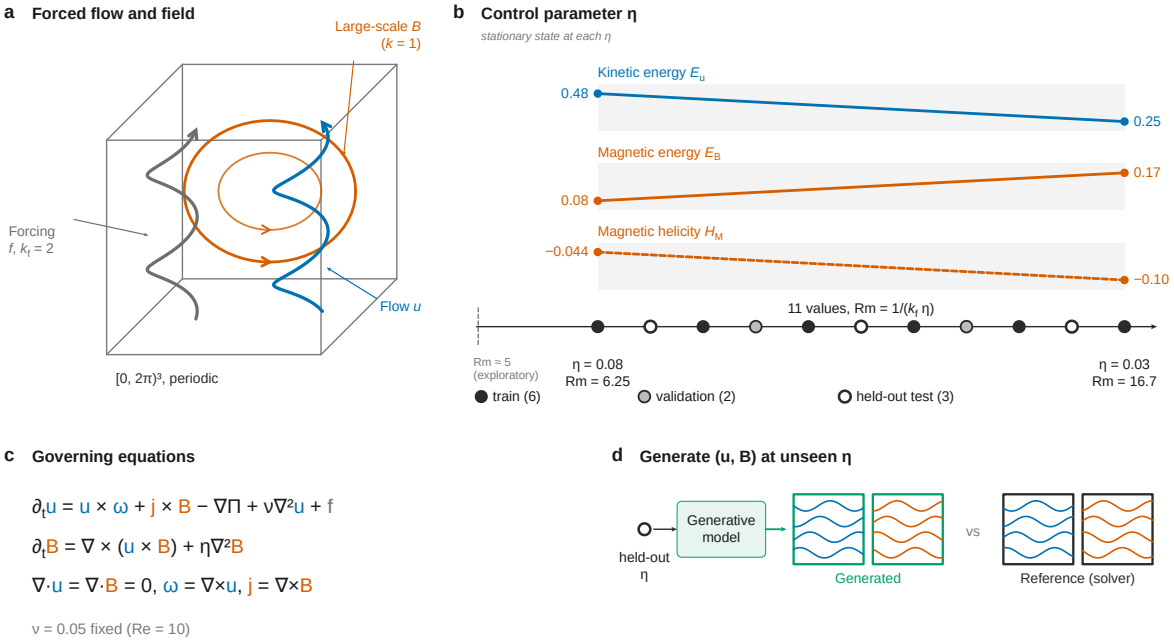


Figure 1.1: The parametric problem of experiment E4. (a) Periodic box with the helical ABC forcing at wavenumber $k_f = 2$, the flow it drives and the large-scale magnetic field at $k = 1$. (b) The eleven values of the magnetic diffusivity η, alternating between training, validation and held-out test values, with the trends of the kinetic energy, magnetic energy and magnetic helicity of the saturated state (end values measured, segments schematic). (c) The equations; the viscosity is fixed. (d) The task: draw snapshots (u, B) at a held-out η and compare their law with that of the solver. Schematic drawn with the `schematic-figure` skill; source in `figures/mhd-dynamo-problem/`.

negative results are kept. Thirty-one hypotheses are in the register, twenty-eight of them in this experiment (H4 to H31): six were registered after external reviews of the first and third versions of this report, and the last four (H28 to H31, registered after the reviews of the fourth) are not yet run.

1.3 Contributions

All of the following are measured in this repository, on the system described above.

1. **A data set and a metric suite for a parametric 3D dynamo** (Experiments): 11 values of η, 24 trajectories each, with a convergence check on finer grids, and metrics covering the four families of the offer (spectra, helicity, divergence, topology), the energy budgets, and two tests that go beyond statistics chosen in advance, a restart test in the solver and a polarity test.
2. **A comparison of constraint mechanisms by the place where they act** (Comparative study): by design (vector potentials, equivariance under $B \to -B$, an explicit affine term, an anchor on a Gaussian law), by the loss (divergence penalty, law-level stationarity identities) and by the sampling procedure (Leray projection, projected flow, exponential tilting, per-sample rescaling), against an unconstrained baseline, a diffusion model and a Gaussian baseline without a network.
3. **A diagnosis of why the first generators fail, a remedy, and its limit** (Results): networks narrower than the latent dimension do worse than the trivial Gaussian velocity field; giving the velocity field its linear part explicitly reduces the distance to the reference law by a factor of about twenty; two Gaussian baselines without a network remain closer to the reference in energy distance, and the second, which keeps the covariance between Fourier modes, has the magnetic production that the first lacks; a split of the production and a Gaussian law fitted to the samples of the best learned model show what that model holds beyond its first two moments (part of a third-order production) and how little this changes its null count and its energy distance.
4. **A stationarity identity without parameters**, $\mathbb{E}[\langle j \cdot B \rangle] = 0$, that follows from the magnetic-helicity budget and can be imposed at parameter values without data (Methods), and the measurement of what a law-level loss built on it and on the energy identity does and does not change.
5. **Code in the project toolbox** (Toolbox): fifteen registered constraint mechanisms with one test each (two of them, the anchor on the full-covariance Gaussian law and the re-colouring of the second moments at sampling time, are tested and not yet used in a full run), a diffusion sampler, a null-point finder, and the tooling to run and account for GPU jobs.

The report also states what was refuted, by the data or by the reviewing roles, including its own first interpretation of the results (Discussion).

1.4 Outline

the section State of the art places the work in the literature. The section Methods defines the system, the generative model, each constraint mechanism and each metric. The section Experiments gives the data and the protocol. The section Results reports the measurements, the section Comparative study gathers them in one table with the verdict of each hypothesis, the section Toolbox describes the code, the section Discussion discusses limits and negative results, and the section Reproducibility lists the commands, commits, GPU jobs and costs.

2 State of the art

Everything in this section is **published** work, cited from the verified source files of the repository (`docs/specs/sources/literature.md`, `dynamo.md`, `data-metrics.md`); the full references are in the section References.

2.1 Diffusion and flow matching

Diffusion models learn to reverse a noising process, either as a discrete chain (Ho, Jain, and Abbeel 2020) or as a stochastic differential equation whose reverse drift involves the score of the noised law (Song et al. 2021). Flow matching (Lipman et al. 2023) and the closely related rectified flows (Liu, Gong, and Liu 2023) and stochastic interpolants (Albergo and Vanden-Eijnden 2023) learn instead the velocity field of an ordinary differential equation that transports a simple law to the data law; with a straight path between a noise sample and a data sample, the regression target is the difference of the two. The parameterisation of a diffusion model by a "velocity" $v = \alpha_t \varepsilon - \sigma_t x$ was introduced by Salimans and Ho (2022) (their Section 4) and is the one used for the diffusion baseline of this report. Both families extend to function spaces (Kerrigan, Migliorini, and Smyth 2024; Lim et al. 2025).

2.2 Generative models of flows and of PDE solutions

Generative models have been applied to turbulent flows to draw turbulent states directly, without rolling out a simulation (Lienen et al. 2024), to PDE solutions in function space with constraints built into the architecture (Wang et al. 2026), to reconstruction from sparse data (Shu, Li, and Barati Farimani 2023; Huang et al. 2024), and to latent representations of flows (Du et al. 2024). FunDiff (Wang et al. 2026) is the main reference of the offer: a generative model over function spaces, with a latent model trained by rectified flow and a decoder that can enforce constraints such as zero divergence through a stream function.

2.3 Three places for a physical constraint

By design. A field written as the curl of a potential is divergence-free whatever the potential; Richter-Powell, Lipman, and Chen (2022) build divergence-free networks on this principle, and FunDiff uses a stream function. Symmetries can also be built in: the flow of an equivariant vector field transports an invariant base law to an invariant law (Köhler, Klein, and Noé 2020).

By the training loss. Physics-informed diffusion models add the residual of the governing equation, evaluated on a denoised estimate, to the training loss (Bastek, Sun, and Kochmann 2025). The estimate is a conditional mean, so for a nonlinear residual the penalty does not vanish at the optimum (the Jensen gap discussed in that paper). Baldan et al. (2026) report a trade-off between the distributional accuracy and the physical accuracy of flow-matching models trained with a residual term.

By the sampling procedure. Constraints can be imposed during sampling by guidance with the gradient of a residual (Chung et al. 2023; Huang et al. 2024), by projection of each iterate onto the constraint set (Christopher, Baek, and Fioretto 2024), or by corrections interleaved with the integration of a flow-matching model, without retraining (Cheng et al. 2025; Utkarsh et al. 2025). For incompressible flows, the Leray projection has been used inside diffusion samplers (Genuist et al. 2026; Li et al. 2026).

2.4 Helical dynamos in periodic boxes

A flow forced with helicity at an intermediate scale in a periodic box generates a strong magnetic field at the scale of the box, and the time needed to build it grows with Rm (Brandenburg 2001); that paper attributes the underlying inverse cascade of magnetic helicity to Pouquet, Frisch, and Léorat (1976). The magnetic helicity $\langle A \cdot B \rangle$ is an invariant of ideal MHD (Woltjer 1958) and evolves, in a periodic domain, as $\mathrm{d}\langle A \cdot B \rangle / \mathrm{d}t = -2\eta \langle j \cdot B \rangle$ [the review of Brandenburg and Subramanian (2005), their Section 3.4, as recorded in the source file; in this work the identity is also checked numerically against the solver by a test]. The forcing used here is the ABC (Arnold–Beltrami–Childress) flow, whose kinematic dynamo properties have been studied in detail (Alexakis 2011; and Galloway and Frisch 1986, known here from its bibliographic record only). Dealiased pseudo-spectral codes are a standard tool for such periodic boxes (for instance Ponty et al. 2005).

2.5 Metrics

The energy distance of Székely and Rizzo (2013) is a distance between two laws computable from samples; it is zero only when the laws are equal. Magnetic null points, where $B = 0$, organise the topology of a magnetic field (Parnell et al. 1996); the trilinear method of Haynes and Parnell (2007) locates them on a grid.

2.6 What was not found

A literature search made for this repository on 6 November 2026 and extended on 12 November found no generative model trained on dynamo simulations, no use of the magnetic-helicity budget as a constraint of a generative model, no use of null-point counts as an evaluation metric of a generative model, and no restart test of generated MHD snapshots. These are negative statements: an absence in a search is not a proof of novelty, and the source files mark them as unconfirmed.

The repository's own theory brief (`docs/specs/theory-brief.md`) gives the propositions used below. Its Propositions 1 to 7 are elementary and probably known in other forms; in particular the use of weak stationarity identities as the loss of a sample generator is published (Cai et al. 2026), and the present work only claims their use for a field PDE, at the level of the law of a parametric conditional model.

3 Methods

3.1 The physical system

The fields are the velocity u and the magnetic field B (in Alfvén units) of an incompressible conducting fluid in the periodic box $[0, 2\pi)^3$:

$$\partial_t u = u \times \omega + j \times B - \nabla \Pi + \nu \nabla^2 u + f, \qquad \partial_t B = \nabla \times (u \times B) + \eta \nabla^2 B, \qquad \nabla \cdot u = \nabla \cdot B = 0, \tag{3.1}$$

with the vorticity $\omega = \nabla \times u$, the current density $j = \nabla \times B$ and the total pressure Π. The forcing is a constant ABC force at wavenumber $k_f = 2$,

$$f = \nu k_f^2 \, (\sin k_f z + \cos k_f y, \ \sin k_f x + \cos k_f z, \ \sin k_f y + \cos k_f x),$$

which is helical ($\nabla \times f = k_f f$) and whose amplitude makes the unit ABC flow the laminar solution without a magnetic field. The viscosity is fixed, $\nu = 0.05$, so that $Re = 1/(k_f \nu) = 10$. The control parameter is the magnetic diffusivity η, between 0.03 and 0.08, that is $Rm = 1/(k_f \eta)$ between 6.2 and 16.7.

The solver is the pseudo-spectral code of the toolbox (`milady.sim.mhd3d`) on a 16^3 grid with the 2/3 rule: the Fourier modes with $|m_i| \le 5$ in each direction are kept. Counting one mode for each pair $(m, -m)$ of a real field, that is $(11^3 - 1)/2 = 665$ independent complex modes.

3.2 Stationarity identities

A statistically stationary state has an invariant law μ_η. For any smooth observable φ of the state, the mean of its time derivative under μ_η is zero (Proposition 5 of the theory brief). Three quadratic observables give identities that involve only quantities computable on one snapshot (Figure 3.1):

$$\begin{aligned} \text{energy:} \quad & \mathbb{E}\big[\langle f \cdot u \rangle - \nu \langle |\omega|^2 \rangle - \eta \langle |j|^2 \rangle\big] = 0, \\ \text{magnetic helicity:} \quad & \mathbb{E}\big[\langle j \cdot B \rangle\big] = 0, \\ \text{cross helicity:} \quad & \mathbb{E}\big[\langle f \cdot B \rangle - (\nu + \eta) \langle \omega \cdot j \rangle\big] = 0, \end{aligned} \tag{3.2}$$

where $\langle \cdot \rangle$ is the spatial mean and $\mathbb{E}$ the mean over the law. The first is the balance between injection and dissipation. The second follows from the magnetic-helicity budget quoted above: in a periodic box the magnetic helicity changes only through $-2\eta \langle j \cdot B \rangle$, so the mean current helicity of a stationary state vanishes. It contains no parameter and no forcing term: it holds at every η, including values without data. The third is odd in B and is satisfied trivially by a law that is invariant under $B \to -B$.

A fourth identity is used as a metric and not imposed: the magnetic energy budget, $\mathbb{E}[P - \eta\langle|j|^2\rangle] = 0$ with the production $P = \langle j \cdot (u \times B)\rangle$, the balance between the work of the flow on the field (the dynamo term) and the Ohmic dissipation. P is cubic in the fields: it is a sum over triads of wavevectors $k+p+q=0$ of third-order moments $\mathbb{E}[\hat{u}(k)\hat{B}(p)\hat{B}(q)]$. For a Gaussian law with zero mean magnetic field these moments reduce exactly to $\mathbb{E}[\hat{u}(k)]\,\mathrm{Cov}(\hat{B}(p), \hat{B}(q))$: the mean production of a Gaussian law is the contraction of its mean flow with the covariance of the magnetic field between different Fourier modes. The constant forcing gives this system a strong mean flow at k_f and statistics that are not homogeneous, so that covariance exists. A law whose Fourier modes are independent has none of it and therefore no mean production, whatever its spectra. A Gaussian law that keeps the covariance between modes has the part of the production carried by second-order statistics; what it cannot have is the third-order cumulant of the true law. This elementary remark separates the two Gaussian baselines used below.

These are constraints on the law, not on each sample: the injection and the dissipation of one snapshot differ, and their difference is the instantaneous $\mathrm{d}E/\mathrm{d}t$. Forcing each sample to satisfy the balance sets the variance of $\mathrm{d}E/\mathrm{d}t$ to zero and cannot give the right law (Corollary 6 of the theory brief).

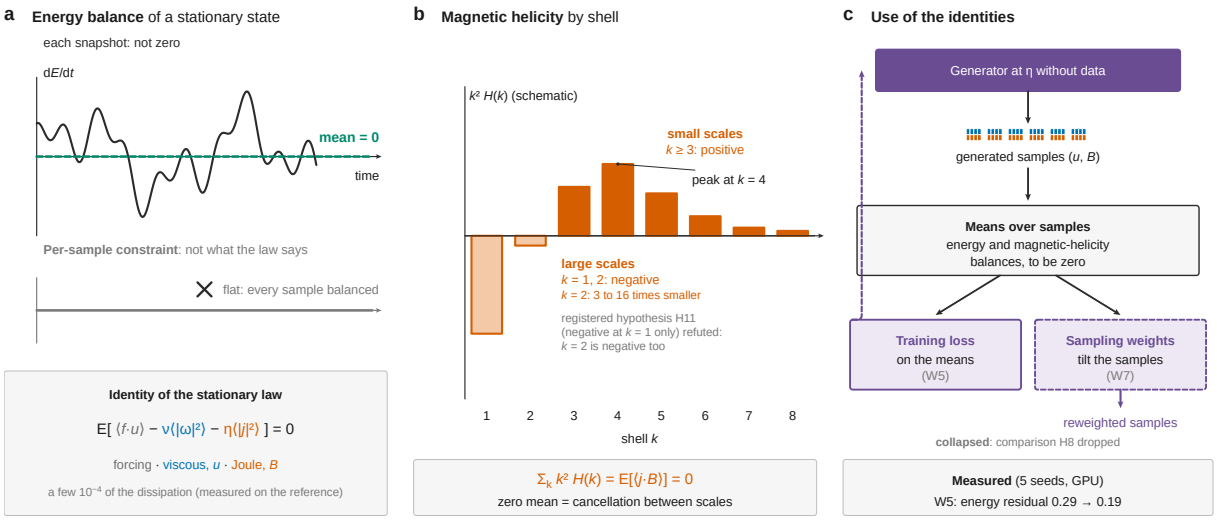


Figure 3.1: The three stationarity identities used in E4. Each is the mean, over the stationary law, of the time derivative of a quadratic invariant of ideal MHD; each term is computable on a single snapshot. The magnetic-helicity identity has no parameter in it. Schematic; source in `figures/stationarity-identities/`.

3.3 Latent representation and generative model

A snapshot (u, B) is transformed to Fourier space and restricted to the 665 modes of the solver. Two representations are used (Figure 3.2).

Free components. The three complex components of $\hat{u}$ and $\hat{B}$ at each mode: $665 \times 12 = 7980$ real numbers. Nothing forces $k \cdot \hat{u} = 0$ or $k \cdot \hat{B} = 0$.

Vector potentials. The potentials in Coulomb gauge, $\hat{A} = \mathrm{i}k \times \hat{B}/|k|^2$ and the same for u, written in the Craya–Herring basis (e_1, e_2) of the plane perpendicular to k: two complex numbers per mode and per field, $665 \times 8 = 5320$ real numbers. Decoding, $\hat{B} = \mathrm{i}k \times \hat{A}$, gives a divergence-free field whatever the latent (Proposition 1 of the theory brief). Because the modes are exactly

those of the solver, every quadratic quantity of Equation 3.2 computed from the latent is the one the solver would compute.

Both latents are normalised mode by mode. The generative model is conditional flow matching on the straight path (Lipman et al. 2023): with $x_0 \sim \mathcal{N}(0, I)$, a data latent x_1 at the condition c (the normalised $\log \eta$) and $x_t = (1-t)x_0 + tx_1$, a network $v_\theta(x, t, c)$ is trained by regression,

$$\mathcal{J}(\theta) = \mathbb{E}\left\|v_\theta(x_t, t, c) - (x_1 - x_0)\right\|^2, \tag{3.3}$$

and a sample is obtained by integrating $\mathrm{d}x/\mathrm{d}t = v_\theta(x, t, c)$ from $t = 0$ to 1 with Heun's scheme (50 steps). The network is a multilayer perceptron acting on the flattened latent: a linear map from the latent to a hidden space of width 512, four residual blocks in that space, and a linear map back to the latent, with Fourier embeddings of t and c.

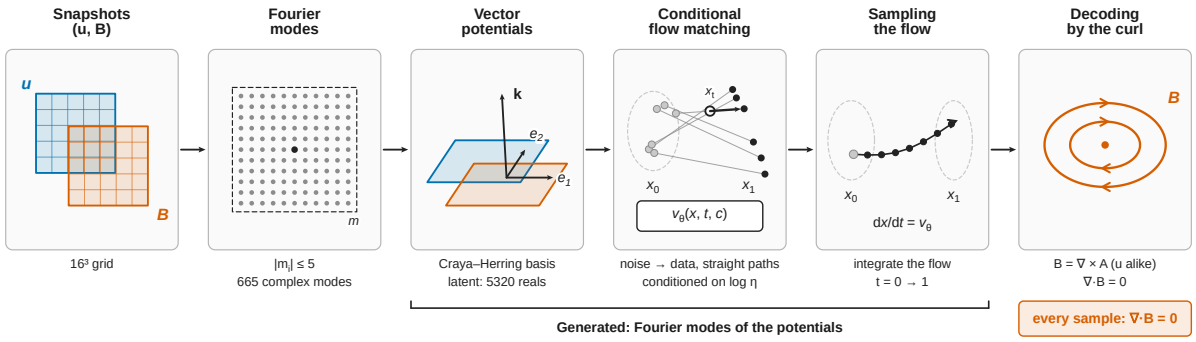


Figure 3.2: The generative pipeline: snapshots, Fourier modes of the solver, vector potentials in the Craya–Herring basis, conditional flow matching, sampling, decoding. Every decoded sample is divergence-free. Schematic; source in `figures/generative-pipeline/`.

3.4 Constraint mechanisms

The mechanisms are grouped by the place where they act (Figure 3.3); the names W0 to W11 and G0 are those of the result files.

Baseline (W0). Flow matching on the free components, no constraint.

Model design.

- *Vector potentials* (W4): flow matching on the potential latent.
- *Equivariance under* $B \to -B$ (W4s). Equation 3.1 is invariant under $B \to -B$, so the stationary law is too (when it is unique). With S the map that changes the sign of the magnetic channels, the field $v_{\mathrm{sym}}(x) = \frac{1}{2}\left[v(x) + Sv(Sx)\right]$ is equivariant and the generated law is exactly invariant (Proposition 4 of the theory brief; Köhler, Klein, and Noé (2020)).
- *Explicit affine term* (W9): $v(x, t, c) = G(t, c)\, x + b(t, c) + \mathrm{network}(x, t, c)$, with one 8×8 matrix G and one offset b per Fourier mode, both read linearly from an embedding of (t, c) and initialised at zero. The reason for this term is given in Section 3.5. W9 also uses the smooth embedding of the condition of W4c (frequencies up to half a cycle over the range of $\log \eta$), so it differs from W4s in two ways; the matched control, the same network with the smooth embedding and no affine term, is the variant of width 512 of the width scan.

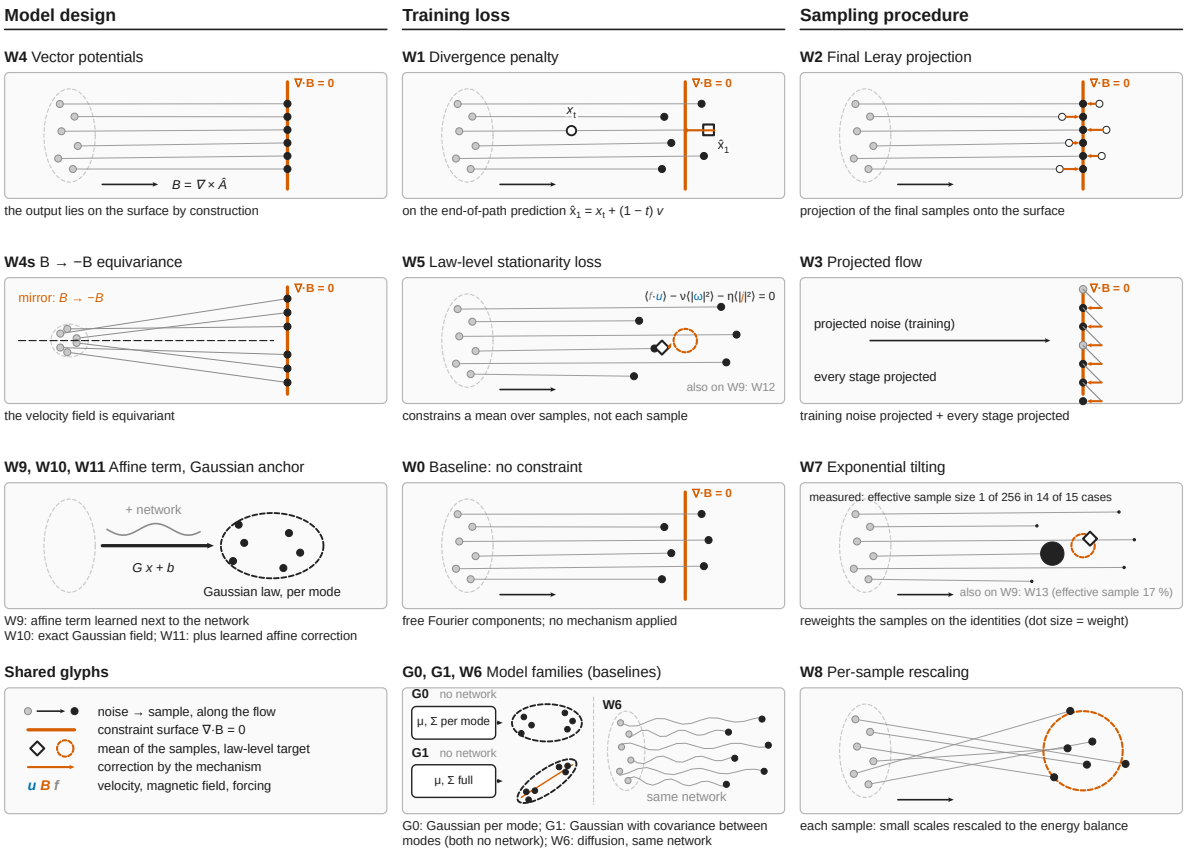


Figure 3.3: The constraint mechanisms compared in E4, by the place where they act: model design, training loss, sampling procedure. Dot positions are illustrative; no result is shown. Schematic; source in `figures/constraint-mechanisms/`.

- *Gaussian anchor* (W10) and *anchor with affine correction* (W11): $v = v_{\text{Gauss}}(x, t, c) +$ network, where v_{Gauss} is the exact flow-matching field of a mode-wise Gaussian law fitted to the data (Equation 3.4); W11 adds the learnable affine term of W9 to the correction.

Training loss.

- *Divergence penalty* (W1): on the free components, the loss Equation 3.3 plus 10 times the squared relative divergence of the end-of-path prediction $\hat{x}_1 = x_t + (1-t)v_\theta$. The constraint is linear, so the penalty vanishes at the optimal field and does not bias it (Proposition 3 of the theory brief).
- *Law-level stationarity loss* (W5): the equivariant potentials model is fine-tuned (400 iterations) with an additional term that draws, at four values of η taken at random in the whole range (with or without data), 16 samples through a differentiable 6-step integration, and penalises the squared normalised means of the energy and magnetic-helicity residuals of Equation 3.2. The square of a mean is estimated without bias by the product of the means of two half-batches; the naive square of one batch mean would add the variance of the residual and penalise the spread of the law. A control (W5_control) receives the same fine-tuning without the term.

Sampling procedure.

- *Final projection* (W2): Leray projection $\hat{f} - k\,(k \cdot \hat{f})/|k|^2$ of the samples of W0.
- *Projected flow* (W3): the noise is projected, the model is trained with projected noise, and every stage of Heun's scheme is projected, so that the whole path stays in the divergence-free subspace (Proposition 2 of the theory brief).
- *Exponential tilting* (W7): the samples of a trained model receive weights $w_i \propto \exp(\lambda \cdot g_i)$, where g_i are the per-sample residuals of the identities and λ is chosen so that the weighted means vanish; this is the law closest to the generated one, in relative entropy, that satisfies the identities (Proposition 7 of the theory brief; the existence and characterisation of this projection are due to Csiszár (1975)).
- *Per-sample rescaling* (W8): the modes above the forcing scale of each sample are multiplied by a factor that makes its own injection equal its own dissipation. This is the per-sample version of a law-level identity, included as the counter-example of Corollary 6.

Loss and sampling on a model close to the reference. The stationarity loss and the tilting were first applied to a model far from the reference (W4s). They were applied again to the model with the affine term: W12 is W9 fine-tuned with the stationarity loss (and a control with the same fine-tuning without the term), and W13 is the tilting of the control samples.

Model families.

- *Diffusion* (W6): the same network and latent as W4, trained by v-prediction on a variance-preserving path (Salimans and Ho 2022) and sampled with the ancestral sampler (100 steps, the same number of network evaluations as 50 Heun steps).
- *Mode-wise Gaussian law* (G0): no network. For each training η, one mean and one 8×8 covariance per Fourier mode are estimated from the data, made invariant under $B \to -B$, and interpolated linearly in $\log \eta$. This law reproduces the second-order statistics of each mode (hence the spectra, the helicities and the mean quadratic budgets) and nothing else; in particular it has no covariance between different modes.
- *Gaussian law with the full covariance* (G1): no network either. For each training η, the mean and the full sample covariance of the 5320-dimensional latent are kept (as a centred data matrix: the covariance has rank at most the number of snapshots), made invariant under $B \to -B$ by adding the image of every snapshot, and interpolated in $\log \eta$ as for G0: a sample at an intermediate η is a Gaussian draw with the convex combination of the

two neighbouring covariances. G1 differs from G0 only by the covariance between modes. It was added after the external review of a first version of this report, to separate what is Gaussian from what is not (Section 5.4). G1 is the point of comparison for any claim that a learned model holds non-Gaussian structure.

3.5 The linear part of the velocity field

Proposition (flow-matching field of a Gaussian law). Let $x_0 \sim \mathcal{N}(0, I)$ and $x_1 \sim \mathcal{N}(\mu, \Sigma)$ be independent and $x_t = (1-t)x_0 + tx_1$. The minimiser of Equation 3.3 is affine in x:

$$v_{\text{Gauss}}(x,t) = \mu + (t\,\Sigma - (1-t)\,I)(t^2\Sigma + (1-t)^2 I)^{-1}(x - t\mu). \tag{3.4}$$

Proof. The minimiser is the conditional mean $\mathbb{E}[x_1 - x_0 \mid x_t = x]$. The pair $(x_1 - x_0, x_t)$ is jointly Gaussian, with $\mathbb{E}[x_1 - x_0] = \mu$, $\mathbb{E}[x_t] = t\mu$, $\mathrm{Cov}(x_1 - x_0, x_t) = t\Sigma - (1-t)I$ and $\mathrm{Cov}(x_t) = t^2\Sigma + (1-t)^2 I$. The conditional mean of a Gaussian vector is the linear regression. □

This is elementary and certainly known. Two consequences matter here. First, for independent standard Gaussian coordinates ($\mu = 0$, $\Sigma = I$) the field is $v = (2t-1)x/(2t^2 - 2t + 1)$ and the minimum of the loss Equation 3.3 per coordinate, averaged over t, is $\int_0^1 \mathrm{d}t/(2t^2 - 2t + 1) = \pi/2 \approx 1.571$: a model whose loss on normalised data is above $\pi/2$ does worse than this trivial field. Second, the network of Section 3.3 maps the latent to a hidden space of width 512 and back, so its Jacobian with respect to x has rank at most 512, whereas the linear part of Equation 3.4 is a full-rank, time-dependent matrix in a latent of 5320 dimensions. That the plain network cannot represent this linear part is exact; that this is the cause of its failure is a **conjecture**, tested by a width scan in the section Results (hypothesis H21). The remedy that does not rely on the conjecture is to give the model the term explicitly (W9) or exactly (W10, W11); Figure 3.4 summarises the argument.

3.6 Metrics

All metrics compare an ensemble of generated snapshots at one η with the reference ensemble at the same η (Figure 3.5).

- **Energy-distance ratio.** Each snapshot is reduced to 21 physical features (logarithms of the shell spectra of u and B for $k = 1, \ldots, 8$, relative kinetic, magnetic and cross helicities, logarithms of the two energies), standardised with the reference. The energy distance (Székely and Rizzo 2013) between 512 generated and 512 reference snapshots is divided by a **noise floor**: the same distance between two reference subsamples of 512 snapshots taken from disjoint halves of the 24 trajectories (mean of 10 splits). The ratio is the number of floors separating the generated law from the reference. A perfect generator would not give 1 but less, because its samples are independent whereas the reference trajectories are correlated in time (an estimate from the two kinds of floors of Figure 4.5 gives about 0.6; it was not measured with a generator); the floor is a scale, not a bound, and it is itself uncertain: its standard deviation over the splits is 35 to 74 % of its mean.
- **Divergence.** $\|\nabla \cdot B\|/\|\nabla B\|$ in the spectral norm, largest value over the ensemble.
- **Energy balance.** $|\mathbb{E}[\langle f \cdot u\rangle - D_\nu - D_\eta]|/\mathbb{E}[D_\nu + D_\eta]$ with the two dissipations $D_\nu = \nu\langle|\omega|^2\rangle$ and $D_\eta = \eta\langle|j|^2\rangle$.

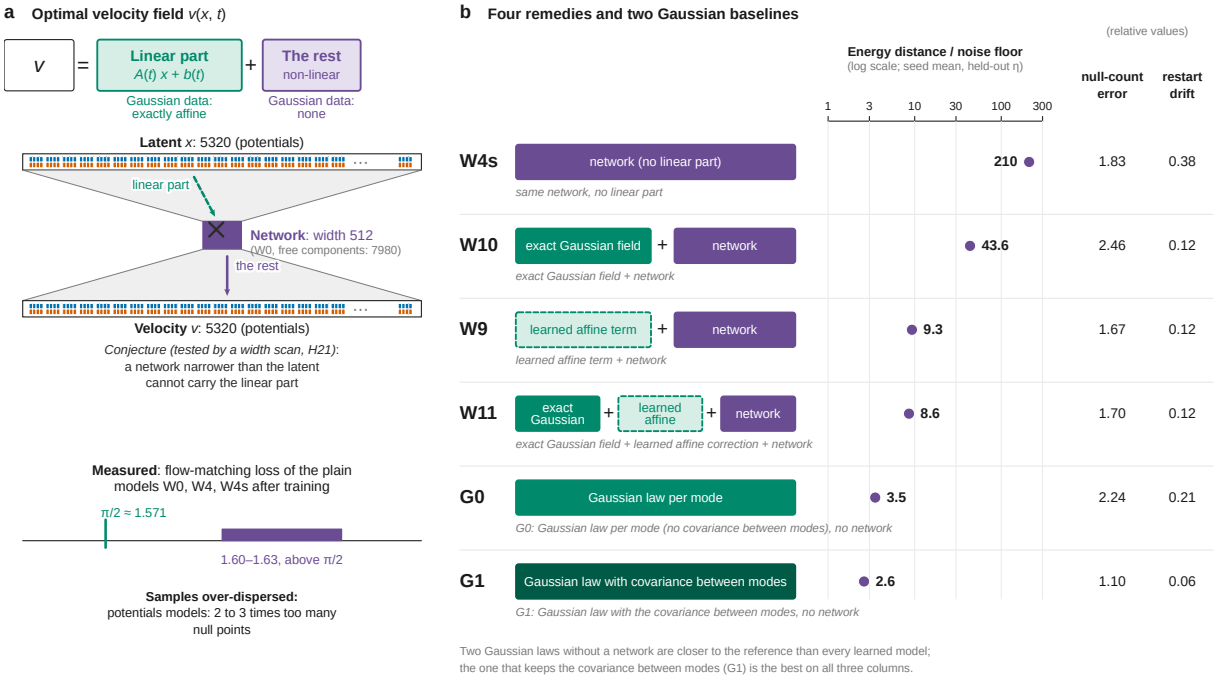


Figure 3.4: Why an explicit linear part. (a) The optimal velocity field of a Gaussian law is affine in x, with a full-rank matrix that depends on time; the plain networks end above the loss of the trivial Gaussian field; the statement on the width is a conjecture. (b) The remedies, the same network without them and the two Gaussian laws without a network: energy distance over the noise floor, null-count error and restart drift of the magnetic energy, seed means over five seeds on the held-out η (measured; values of Table 6.1 and, for G1, Table 5.1). Schematic; source in `figures/linear-part/`.

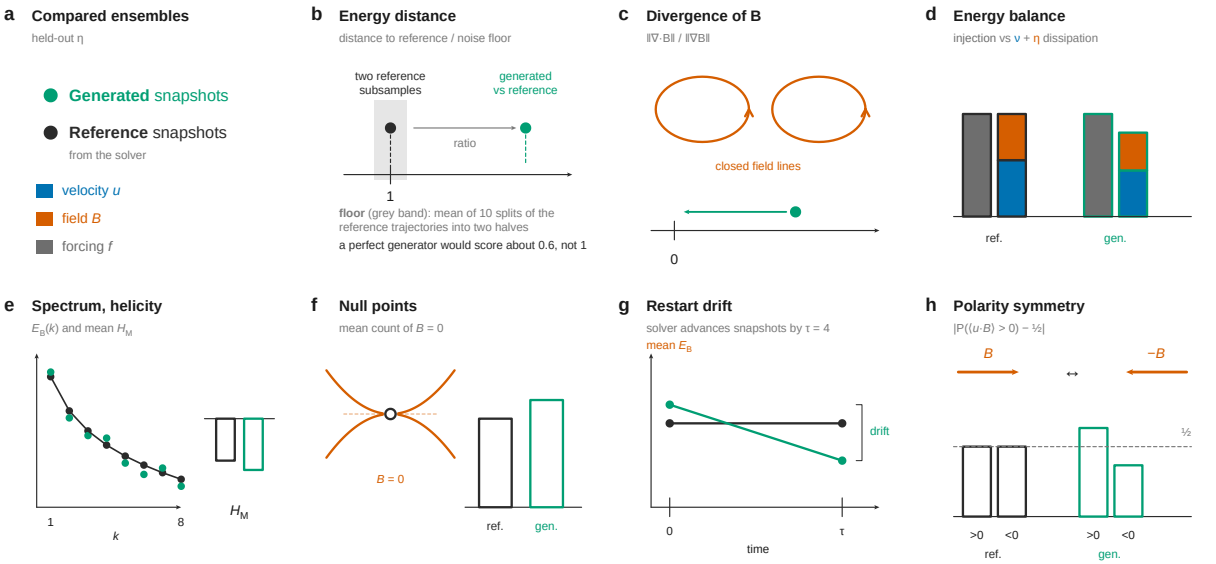


Figure 3.5: The metrics of E4: energy distance over its noise floor, divergence of B, energy balance, spectra and magnetic helicity, magnetic null points, restart drift, polarity. Schematic; source in `figures/metrics/`.

- **Magnetic energy balance.** $|\mathbb{E}[P - D_\eta]|/(|\mathbb{E}[P]| + \mathbb{E}[D_\eta])$ with the production $P = \langle j \cdot (u \times B) \rangle$: zero when the dynamo term balances the Ohmic dissipation, one when there is no production at all.
- **Spectrum and helicity.** Relative L^2 error of the mean shell spectrum $E_B(k)$; relative error of the mean magnetic helicity.
- **Null points.** Relative error of the mean number of magnetic null points per snapshot. The field is evaluated on a 48^3 grid, and the nulls of its trilinear interpolant are found by the method of Haynes and Parnell (2007): cells where the three components change sign, then Newton iterations from nine starting points. On a periodic box the numbers of nulls of the two signs of $\det \nabla B$ must be equal (Poincaré–Hopf), which is used as a test of the implementation.
- **Restart drift.** The generated snapshots are used as initial conditions of the solver and advanced over $\tau = 4$ time units; the metric is the relative change of the ensemble-mean magnetic energy (and, reported with it, of the kinetic energy). It is zero for samples of the true stationary law, and it is the only metric here that tests the generated law against the dynamics rather than against statistics chosen in advance. This is the metric of the objectives, written on the first day, and it has a defect that the external review found late: being relative to the ensemble's own initial energy, it charges a law that starts with a biased energy and relaxes towards the reference for its bias, and a law that starts at the right energy and leaves it for its excursion, without telling the two apart. The last runs therefore also record the course anchored on the reference, $a(\tau) = E_B(\tau)/E_{B,\mathrm{ref}} - 1$ at $\tau = 0.125, 0.25, \ldots, 16$, and its **overshoot** $\max_\tau |a(\tau)| - |a(0)|$: how far the ensemble moves away from the reference beyond its initial deviation (Section 5.5).
- **Polarity.** $|P(\langle u \cdot B \rangle > 0) - \frac{1}{2}|$, zero for a law invariant under $B \to -B$.

4 Experiments

4.1 Reference data

For each of the 11 values of η (logarithmically spaced from 0.08 to 0.03), 24 independent trajectories are started from a small random magnetic seed and a perturbed ABC flow, integrated over 200 time units to let the dynamo saturate, then sampled every 2 time units over 200 time units: 2400 snapshots of (u, B) per value (`experiments/e4_data.py`). Table 4.1 gives the statistics of the saturated state, all **measured**: the kinetic energy falls and the magnetic energy rises as η decreases; the magnetic helicity is negative and its magnitude grows; about 40 % of the magnetic energy sits in the largest scale $k = 1$; the number of magnetic null points per snapshot more than doubles across the range (Figure 4.1).

Table 4.1: Reference data of E4 (measured; `experiments/results/e4_data.json` and `e4_reference.json`). E_u, E_B: kinetic and magnetic energies; $H_M = \langle A \cdot B \rangle$; null points per snapshot (mean $\pm$ standard error over 400 snapshots); $k_\lambda = (\langle |j|^2 \rangle / \langle |B|^2 \rangle)^{1/2}$; energy residual: mean of injection minus dissipation, over the mean dissipation; floor: energy distance between two halves of the reference trajectories.

η	Rm	Split	E_u	E_B	H_M	$E_B(k{=}1)/E_B$	Null points	k_λ	Energy residual	Floor of the energy distance
0.0800	6.2	training	0.491	0.0783	-0.0433	0.436	21.1 $\pm$ 0.47	2.5	-3.0×10^{-5}	0.117
0.0725	6.9	held-out test	0.44	0.0967	-0.056	0.434	21.4 $\pm$ 0.53	2.53	4.4×10^{-6}	0.0703
0.0658	7.6	training	0.402	0.111	-0.0646	0.426	21.9 $\pm$ 0.49	2.57	-4.1×10^{-4}	0.0363
0.0596	8.4	validation	0.373	0.121	-0.0719	0.422	24.2 $\pm$ 0.56	2.61	1.3×10^{-4}	0.0425
0.0540	9.3	training	0.345	0.131	-0.0778	0.413	24 $\pm$ 0.62	2.65	2.2×10^{-4}	0.037
0.0490	10.2	held-out test	0.321	0.14	-0.0841	0.408	27.6 $\pm$ 0.65	2.69	-4.6×10^{-5}	0.0394

η	Rm	Split	E_u	E_B	H_M	$E_B(k{=}1)/E_B$	Null points	k_λ	Energy residual	Floor of the energy distance
0.0444	11.3	training	0.309	0.145	-0.0849	0.396	31.7 ± 0.76	2.74	-2.7×10^{-4}	0.0409
0.0403	12.4	validation	0.288	0.153	-0.0904	0.389	32.5 ± 0.77	2.79	-2.1×10^{-4}	0.0514
0.0365	13.7	training	0.276	0.16	-0.0947	0.384	38.1 ± 0.98	2.84	6.6×10^{-6}	0.0479
0.0331	15.1	held-out test	0.265	0.165	-0.0962	0.371	44.1 ± 1.1	2.9	1.7×10^{-4}	0.0733
0.0300	16.7	training	0.253	0.172	-0.0968	0.364	52 ± 1.2	2.95	2.3×10^{-4}	0.04

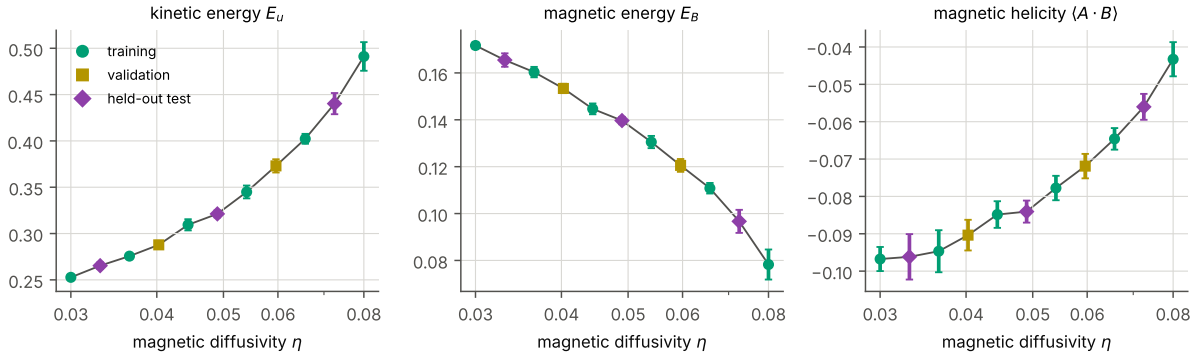


Figure 4.1: Reference data: kinetic energy, magnetic energy and magnetic helicity against η (mean over 24 trajectories; bars: ± 2 standard errors), with the training, validation and held-out values. Data plot from `experiments/results/e4_data.json` (`experiments/e4_plots.py`).

Two properties of the reference were checked before any model was trained.

The stationarity identities hold on the data. The energy residual of Table 4.1 is at most a few 10^{-4} of the dissipation at every η. The mean current helicity $\langle j \cdot B \rangle$ is zero within its standard error: its shell spectrum changes sign, negative at $k \leq 2$ and positive for $k \geq 3$ (Figure 4.2). A hypothesis registered beforehand (H11) predicted a simpler picture, a magnetic helicity spectrum negative at $k = 1$ only and positive for every $k \geq 2$; it was **refuted** by the data, the shell $k = 2$ being negative as well.

The data are converged for the purpose. The same statistics were recomputed at the three held-out η with half the time step and on a 24^3 grid, and at one η on a 32^3 grid (Figure 4.3): energies and helicity agree with those of the 16^3 data within one to two standard errors. The flow at

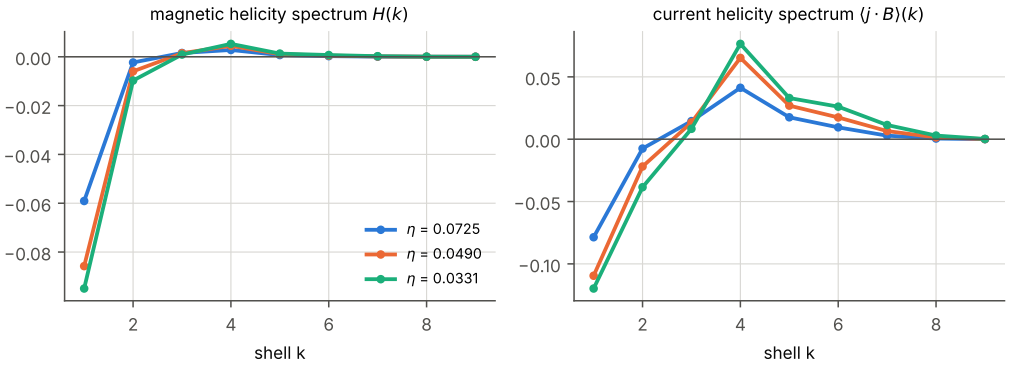


Figure 4.2: Reference data: shell spectra of the magnetic helicity and of the current helicity $\langle j \cdot B \rangle$ at the three held-out η. The mean of $\langle j \cdot B \rangle$ is zero: the negative shells $k \leq 2$ cancel the positive shells $k \geq 3$. Data plot from `experiments/results/e4_reference.json`.

$Re = 10$ is smooth; the small grid is a choice of scale for the comparison, not an under-resolved simulation.

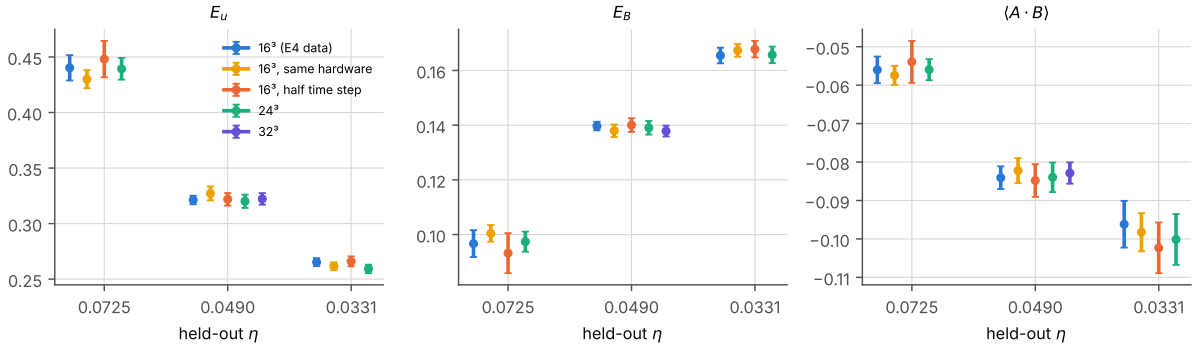


Figure 4.3: Convergence of the reference data: energies and magnetic helicity at the three held-out η for the 16^3 grid of E4, the same grid recomputed on the hardware of the check, half its time step, 24^3 and 32^3 (one η); bars: ± 2 standard errors over 24 trajectories. Data plot from `experiments/results/replications/e4_convergence/`.

4.2 Protocol

Split of the parameter values. The 11 values alternate (Figure 4.4): six training values (the two ends and every second value), two validation values and three held-out test values. The test values are used neither to train nor to choose a model. Every number quoted "on the held-out η" is the mean over the three test values.

Seeds. Each variant is trained with five seeds. A comparison between two variants is the paired difference per seed, quoted as mean $\pm$ standard error over the five seeds (rule 4 of `AGENTS.md`).

Noise floor. The floor of the energy distance (Section 3.6) is computed per η from disjoint halves of the trajectories and not from a random split of the snapshots: snapshots of one trajectory are correlated in time, and a random split gives a floor three to six times lower that no independent run of the solver would reach (Figure 4.5).

Training and sampling. 4000 iterations of Adam, batch 128, learning rate 5×10^{-4}, exponential moving average of the weights; 512 samples per validation and test value, 256 per training value.

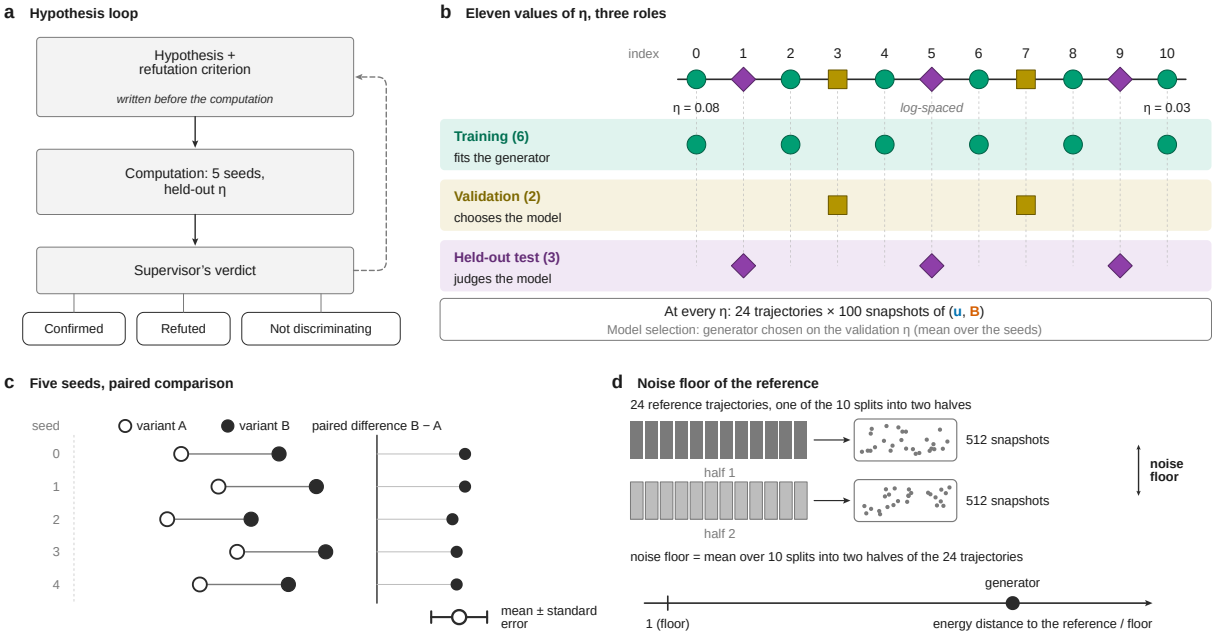


Figure 4.4: The protocol of E4: values of η for training, validation and held-out test; five model seeds; noise floor of the energy distance from two halves of the reference trajectories; hypotheses registered before the computation and reviewed afterwards. Schematic; source in `figures/protocol/`.

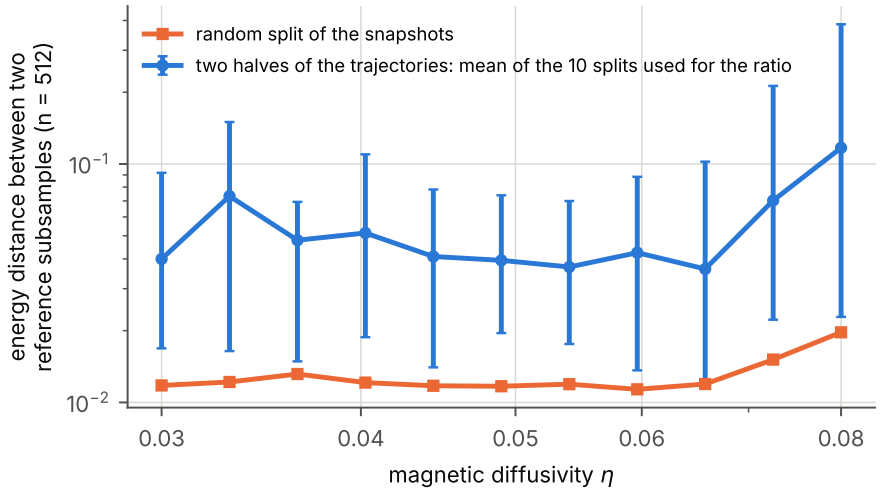


Figure 4.5: Noise floors of the energy distance on the reference data: between two subsamples from disjoint halves of the 24 trajectories (mean of the 10 splits that define the floor; bars: smallest and largest of 20 splits) and between two random subsamples. Data plot from `experiments/results/e4_floor.json`.

The “long” variants use 8 times the iterations. The null points are counted on 200 samples per value and the restart test uses 128 samples.

Register and review. Each hypothesis is written in `research/hypotheses.json` with its refutation criterion before the corresponding run, and the commit of the registration precedes the commit of the results. The `encadrant` role then recomputes the comparison from the result files and gives one verdict per criterion; its reports are in `research/reviews/`. The selection of the generator that is measured against the objectives of the race uses the validation values only.

4.3 Compute

The session runs in a cloud container with four CPU cores and no GPU. The data, the references and the first two seeds were computed there; the five-seed runs were made on rented GPU pods through `tools/runpod_job.py`, which clones the pushed commit on the pod, runs the command under a time limit, brings the outputs back, deletes the pod and writes the cost in `research/compute-ledger.jsonl` (Figure 4.6). A training of 4000 iterations takes 30 s to 2 minutes on the GPU, depending on the variant and on how many seeds share the card, against about 15 minutes on one CPU core; the evaluation (null points, restart test) dominates the run time. The costs are in the section Reproducibility.

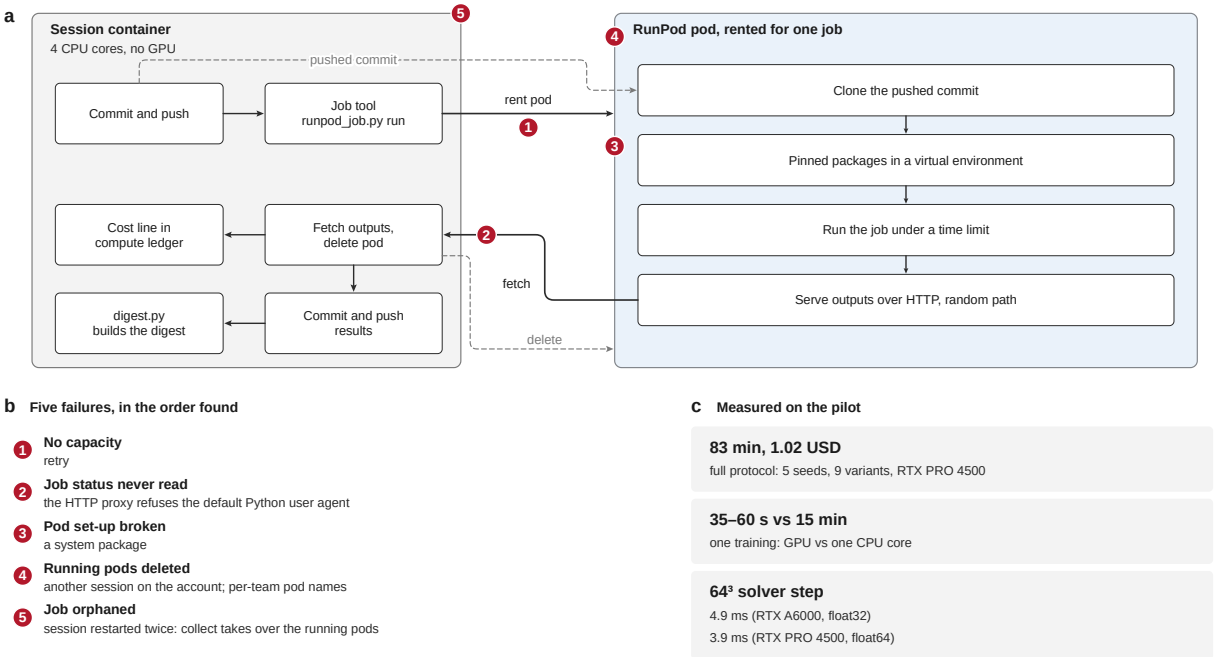


Figure 4.6: The compute path of the pilot: session, GPU pod, results, ledger, and the failures met and repaired on the way. Schematic; source in `figures/compute-path/`.

The workflow of the experiment and the organisation of the agents are drawn as two explorable diagrams, experiment-workflow.html and agent-organisation.html; their static exports are Figure 4.7 and Figure 4.8.

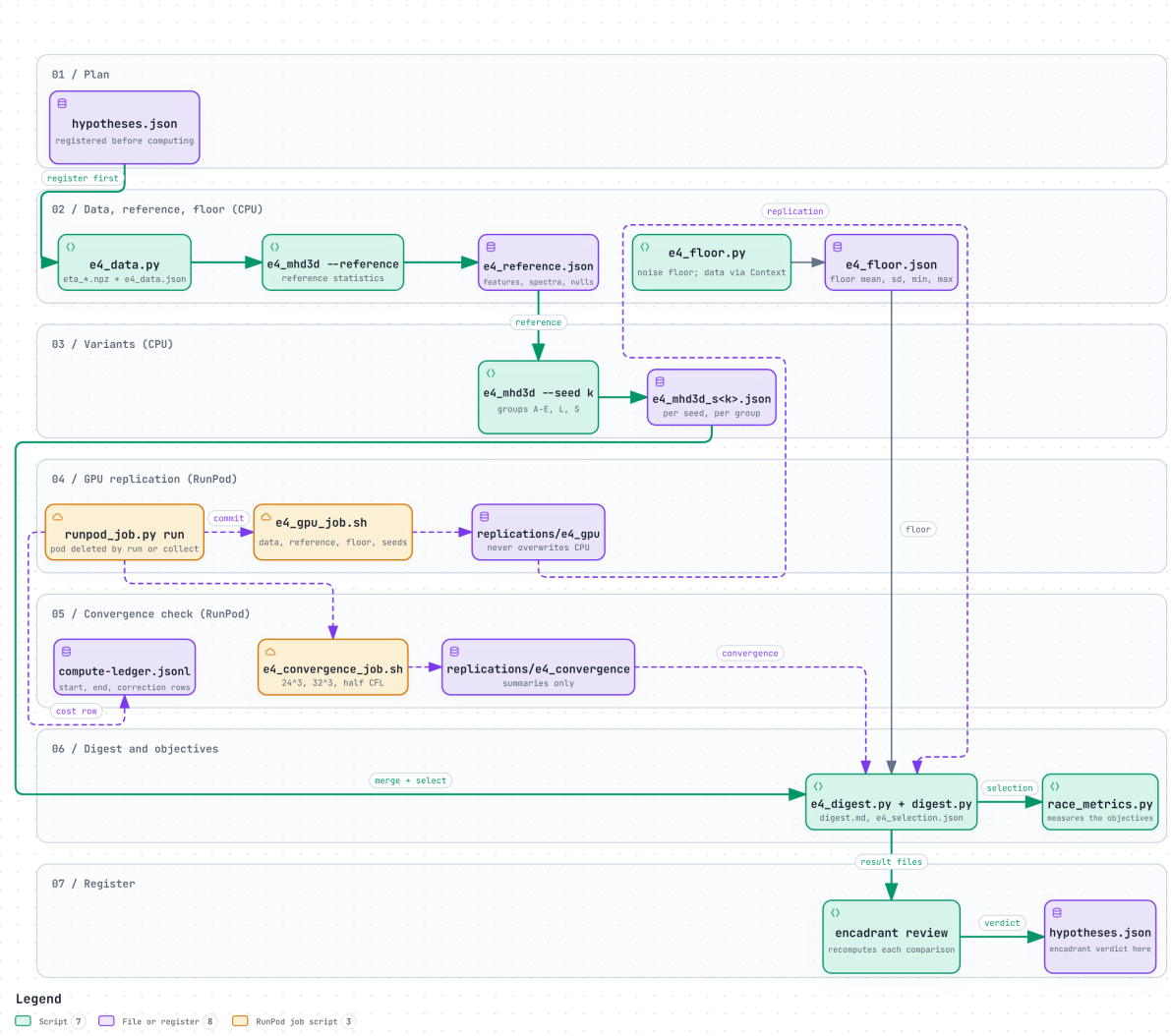


Figure 4.7: Workflow of experiment E4: plan, data, reference and floors, variants, GPU runs, digest, objectives and register. Static export of the Archify diagram `diagrams/experiment-workflow.html`.

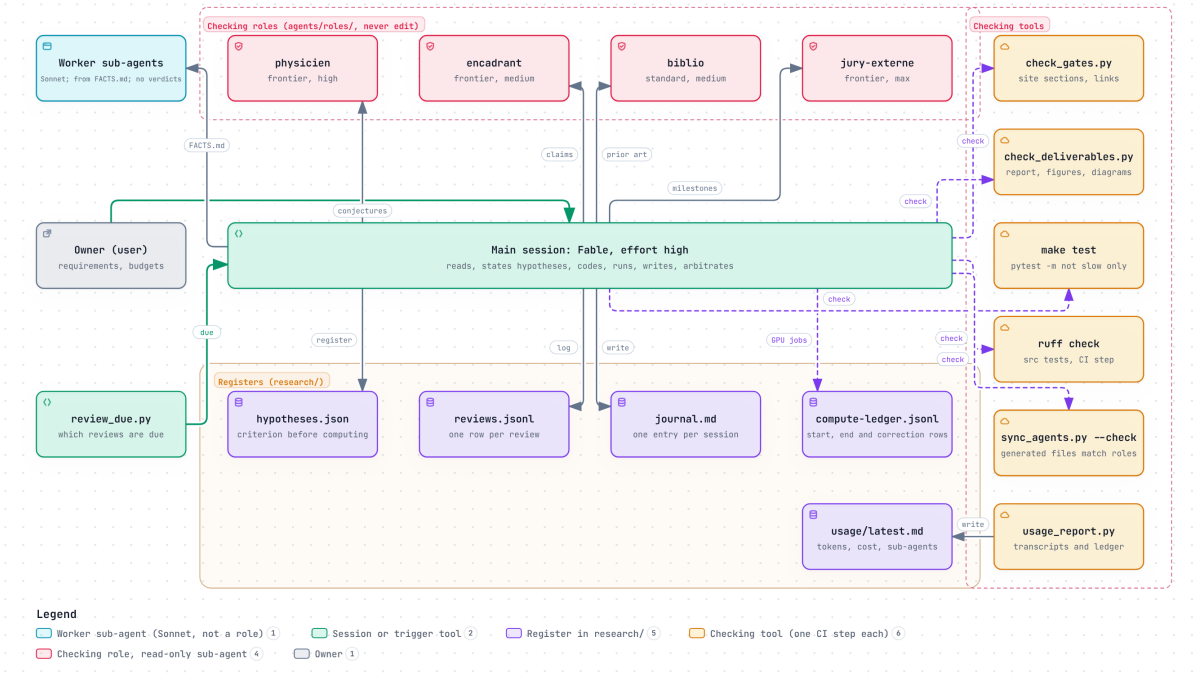


Figure 4.8: Organisation of the team: the main session, the four roles with their model tier and reasoning level, the reviews named by `tools/review_due.py`, the registers and the checking tools. Static export of the Archify diagram `diagrams/agent-organisation.html`.

5 Results

Everything in this section is **measured**, on five model seeds unless a table says otherwise; the numbers are in Table 6.1 of the section Comparative study and in the tables below.

5.1 The registered protocol: every first-generation model is far from the reference

The first protocol, registered before any training, compared the baseline W0, the mechanisms W1 to W5, the diffusion model W6 and the sampling-time corrections W7 and W8, all with the same residual network of width 512 (Figure 5.1).

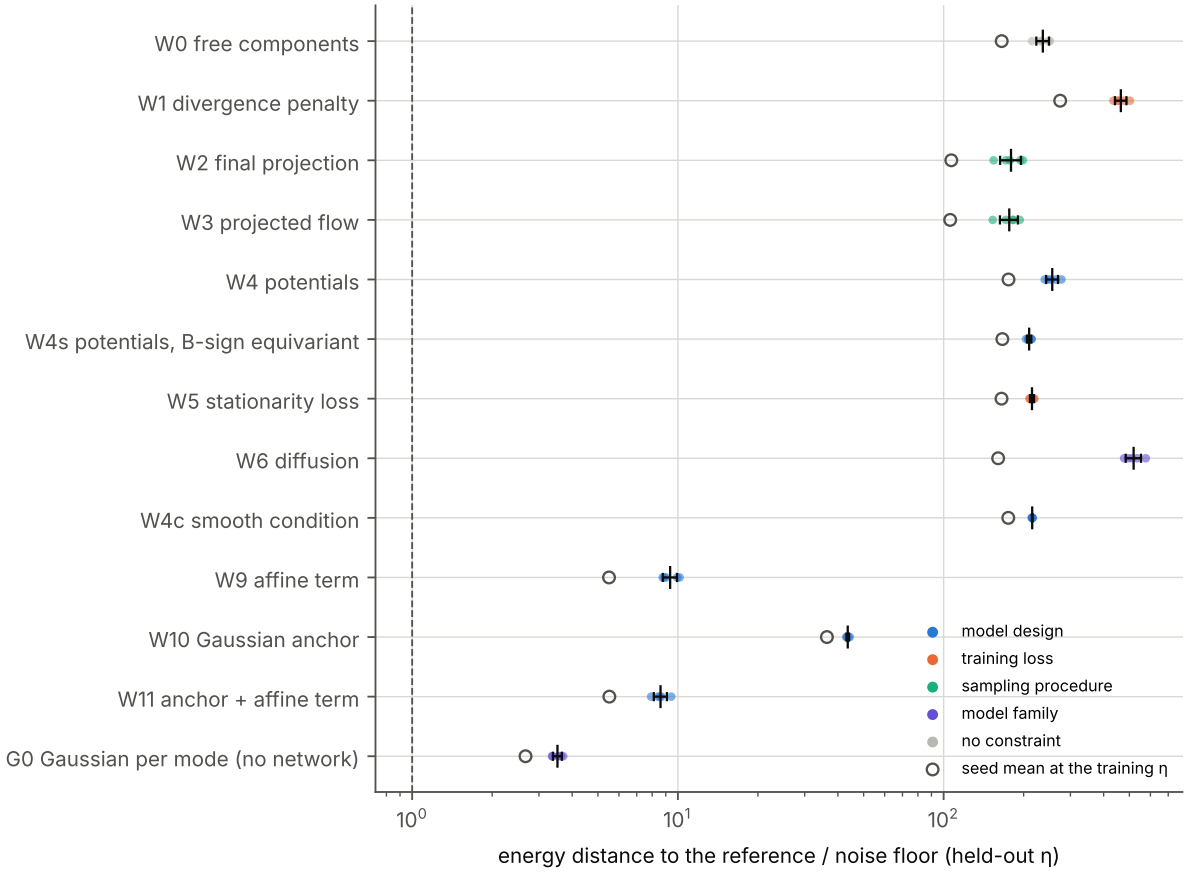


Figure 5.1: Energy distance to the reference divided by the noise floor on the held-out η, per variant: one dot per seed, bar: seed mean ± 2 standard errors (drawn with at least five seeds), open circle: seed mean at the training η; dashed line: the floor. Data plot from `experiments/results/e4_mhd3d_s<k>.json`.

The outcome is negative for all of them: the generated laws are between about one hundred and five hundred floors away from the reference on the held-out values, and not much closer on the

training values. Within this failure the mechanisms can still be told apart, and the registered hypotheses were decided as follows (verdicts of the `encadrant`, Table 6.2).

- **Divergence.** The potentials (W4), the final projection (W2) and the projected flow (W3) bring the relative divergence of B to round-off, about 10^{-8}; the free model stays at order one and the penalty (W1) only halves it (H5 confirmed). The divergence of the free model does not change when the number of integration steps is multiplied by four, from 50 to 200 (H14 confirmed): it is an error of the learned field, not of the integrator, as Proposition 2 of the theory brief predicts for a field that lacks the contraction towards the constraint set.
- **Potentials against free components.** Generating potentials does not bring the law closer to the reference than generating free components: the paired difference has the wrong sign in all five seeds (H4 refuted). The projected flow is not better than a projection of the final samples (H6 refuted); the two are the best of this group.
- **The penalty costs accuracy.** The model trained with the divergence penalty is about twice as far from the reference as the baseline. The penalty is unbiased at the optimum, but the network is far from the optimum, and there the penalty competes with the regression.
- **Stationarity loss.** Fine-tuning with the law-level identities lowers the residual of the mean energy balance by about a third against the control that receives the same fine-tuning without the term (H7 confirmed), at parameter values where no data were used. This is the effect on the energy identity only. The residual of the magnetic-helicity identity, the second term of the loss, does not move (-0.004 ± 0.009), and the loss has side effects that the registered criterion did not look at: against the control, the residual of the magnetic energy balance doubles (0.25 to 0.52), the error on the null count rises by 0.56, the error on the magnetic spectrum falls by 0.56, and the distance to the reference law is unchanged. The comparison between imposing the identities in the loss and by tilting was dropped as not discriminating (H8): the tilted ensemble is one weighted sample, and the registered criterion could be met by its sampling noise.
- **Which identity constrains more.** The registered conjecture that the magnetic-helicity identity is violated more than the energy identity was refuted (H12): on the samples of the equivariant model the normalised residual of $\mathbb{E}[\langle j \cdot B \rangle] = 0$ is about half that of the energy balance.
- **Equivariance.** The polarity asymmetry of the non-equivariant model is already at the sampling noise of 512 samples; the equivariant model cannot lower it (H9 refuted on its first clause). The symmetry is exact by construction in W4s, but in this system it is not what the unconstrained model gets wrong.
- **Flow matching against diffusion.** At the same number of network evaluations, flow matching is about twice closer to the reference than the diffusion model on the held-out η (H10 confirmed as registered). On the training η the order is reversed: the diffusion model is the closer one (160 against 175 floors, in every seed). The advantage of flow matching is in the interpolation in η (Figure 5.2), not in the family at a fixed parameter value.
- **Per-sample rescaling.** W8 satisfies the energy balance of each sample exactly, and has the lowest distance of this group; its law is still about a hundred floors from the reference, and by construction the variance of $\mathrm{d}E/\mathrm{d}t$ of its samples is zero.

5.2 Diagnosis: the models do worse than a Gaussian field

Three measurements point to the same cause. The flow-matching loss of the plain flow-matching models (W0, W4, W4s and the control of W5; the other variants have another loss or another noise and are not comparable), evaluated on training samples, ends between 1.60 and 1.63,

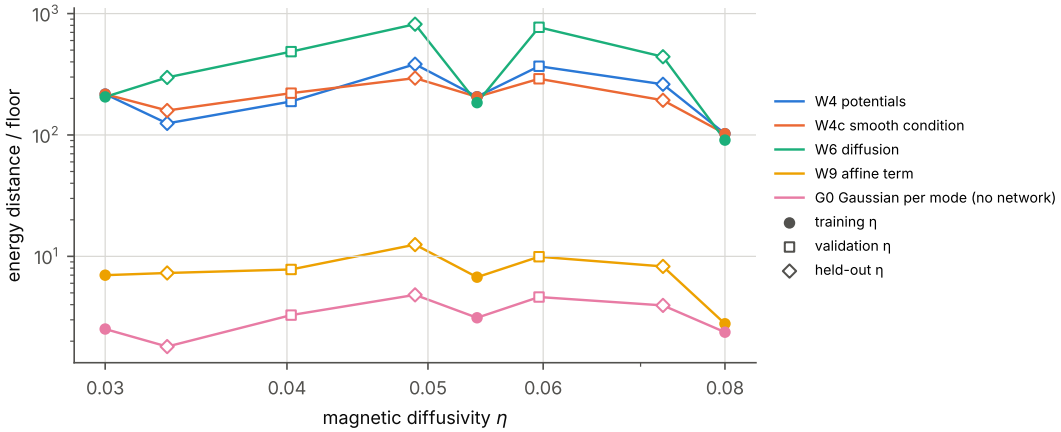


Figure 5.2: Energy-distance ratio against η, seed mean: at the training values (filled circles) and between them (open squares: validation, open diamonds: held-out test). Data plot from `experiments/results/e4_mhd3d_s<k>.json`.

above $\pi/2 \approx 1.571$, the loss of the exact field for independent standard Gaussian coordinates (Section 3.5): after training, these networks fit the normalised data worse than a field that ignores every correlation. The samples of the potentials models are over-dispersed: about two to three times too many magnetic null points (the penalty model W1 and the diffusion model W6 err the other way, with too few) and a fluctuation of the energy budget several times that of the reference. The losses quoted here come from the training logs of the first runs (seeds 0 to 2); the result files store the final loss only for the later variants. And the error is of the same order on the training values as on the held-out ones, so it is not an error of interpolation in η.

Two explanations were registered and tested separately.

The embedding of the condition (H15). The Fourier embedding of the condition used frequencies of up to 8 cycles over the range of $\log \eta$, above what six training values can resolve. A model with a smooth embedding (W4c, frequencies up to half a cycle) was compared with W4.

The linear part of the field (H16, H17). The Gaussian baseline G0 has no network at all; the affine term of W9 gives the network of W4s the linear part of the field explicitly.

The measurements (five seeds, Table 6.1; paired differences in the digest; verdicts of the `encadrant` in Table 6.2) separate the two explanations.

- *The embedding matters for interpolation only.* With the smooth embedding the ratio on the held-out η falls from 256 to 215 (paired difference 40.9 ± 6.8, positive in the five seeds), and on the training η nothing changes (175 for both). The registered criterion asked for a factor of two: H15 is **refuted**, the embedding is a real but minor part of the error.
- *A Gaussian law without a network is fifty times closer.* G0 is at 3.5 floors on the held-out η, against 177 for the best trained model of the first protocol, the projected flow W3 (H16 **confirmed**).
- *The affine term is what was missing.* With the explicit mode-wise affine term (W9) the equivariant network goes from 210 floors (W4s, the comparison registered for H17, **confirmed**) to 9.3. W4s also differs from W9 by its embedding of the condition; against the matched control of the width scan (205 floors, Section 5.7) the factor is 22, in every seed, and the flow-matching loss on training samples falls from 1.61 to 1.18, below $\pi/2$.

5.3 Giving the field its linear part

Four ways of giving the velocity field its linear part were compared (Table 6.1, Figure 5.1, Figure 5.3, Figure 5.4).

- **Sampling the mode-wise Gaussian law itself (G0)** is closer to the reference than any learned model in energy distance (3.5 floors), with a residual of the total energy balance of 0.0015 and spectra within a few per cent. It is also an illustration of what a distance built on second-order features of single modes does not see. Its magnetic energy balance is violated completely (residual 0.998): the law has the right Ohmic dissipation and no production, as the remark of Section 3.2 predicts for independent Fourier modes. It has more than three times the reference number of magnetic null points (relative error 2.24). And when the solver restarts from its snapshots the energies move 15 to 20 % away from their stationary values, against less than 1 % for reference snapshots.
- **Learning the affine term next to the network (W9)** gives 9.3 floors, and **anchoring the field on the exact mode-wise Gaussian field with a learned affine correction (W11)** gives 8.6; the two are indistinguishable within the seed noise. Both are about 2.5 times farther from the reference than G0 in energy distance, with a residual of the total energy balance of 0.05 to 0.06, a magnetic spectrum within 15 % and a magnetic helicity within 20 %. But they have what G0 lacks: the residual of the magnetic energy balance is 0.055 and 0.060 (the first-generation models are at 0.15 to 0.5), the error on the null count falls from 2.24 to 1.7, and the restart drift of the magnetic energy at $\tau = 4$ from 0.21 to 0.12 (42 % less), in every seed; what this single-time drift measures for each law is taken apart in Section 5.5.
- **Anchoring on the exact mode-wise Gaussian field with a plain network (W10)** is worse than the Gaussian law it starts from: 43.6 floors (H19 refuted). The registered explanation (H20) was that the network's own linear response spoils the second-order statistics that the anchor had right, and that a learnable affine correction would restore them. The correction does bring the model from 43.6 to 8.6 floors, but not below G0, and H20 was refuted on that clause; its two other clauses, on the restart drift and the null count, are met.

A first version of this report concluded here that the flow-matching models with an affine term had learned "the third-order correlations that make the field a dynamo". The external review refuted that reading, and the next section reports the measurement that replaces it.

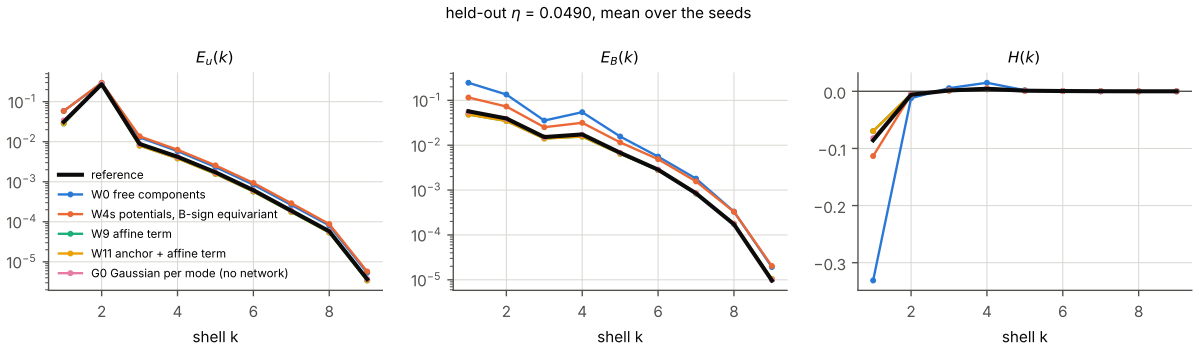


Figure 5.3: Shell spectra of the velocity and of the magnetic field and magnetic helicity spectrum at one held-out η: reference (black) and generated ensembles (mean over the seeds). Data plot from `experiments/results/e4_mhd3d_s<k>.json` and `e4_reference.json`.

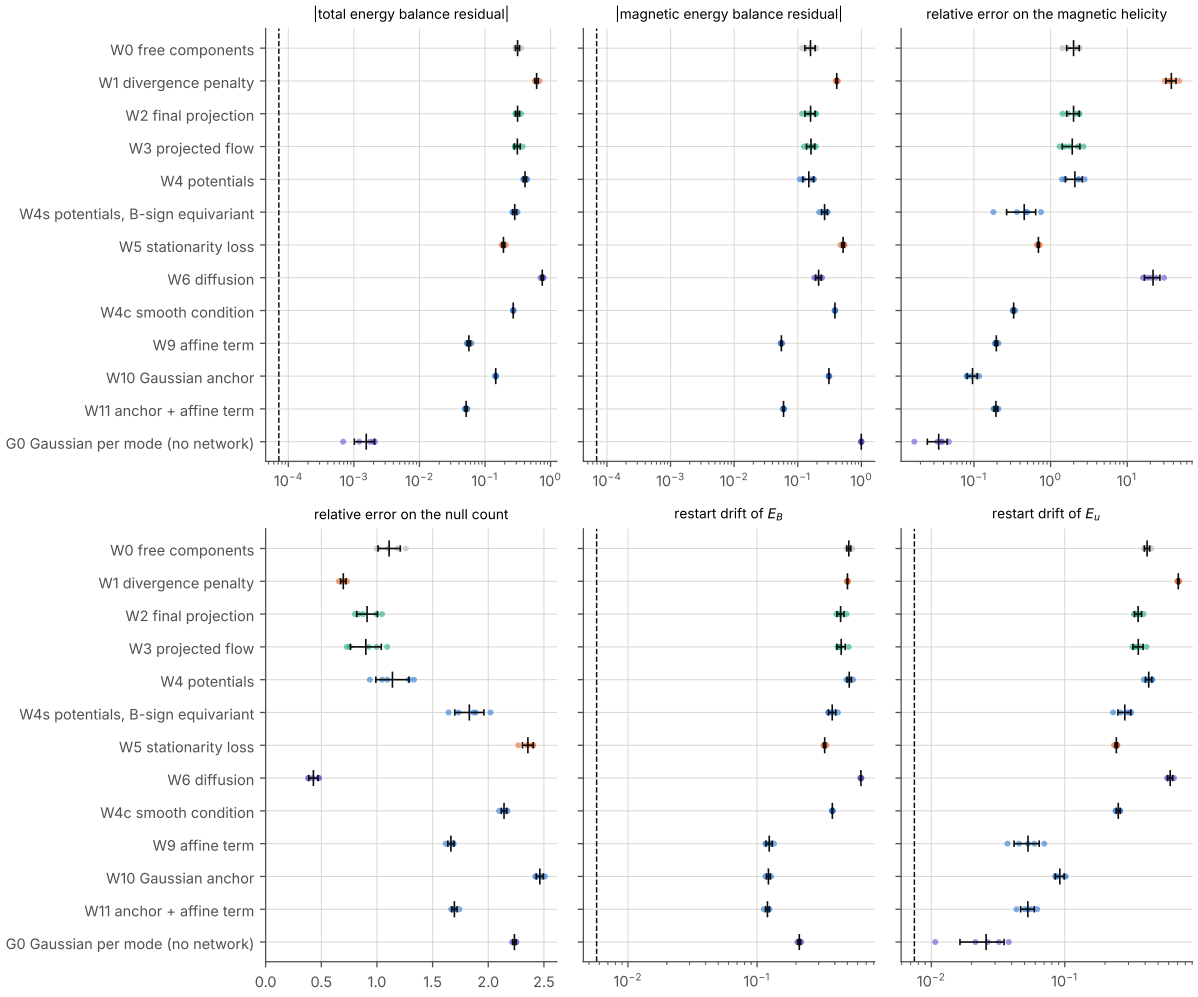


Figure 5.4: Physical metrics on the held-out η, one dot per seed and seed mean ± 2 standard errors: residuals of the mean total and magnetic energy balances, errors on the mean magnetic helicity and on the number of magnetic null points, drifts of the magnetic and kinetic energies when the solver restarts from the generated snapshots. Dashed lines: the same quantity on reference snapshots. Data plot from `experiments/results/e4_mhd3d_s<k>.json`.

5.4 Two Gaussian laws: what the covariance between modes holds

What do W9 and W11 have that G0 lacks? The remark of Section 3.2 gives a candidate that is not third-order: for a Gaussian law, the mean magnetic production is the product of the mean flow and the covariance of B between different modes, and G0 has no such covariance. The hypothesis registered after the review (H22) is that a Gaussian law that keeps this covariance, G1, fitted on the training η only, has a magnetic energy balance residual below 0.2 and beats W9 on the restart drift of the magnetic energy at $\tau = 4$ and on the null count. Table 5.1 gives the measurement, from a job in which G0, W9 and W11 were retrained next to G1 (they reproduce the main run).

H22 is **confirmed**, its three clauses in every seed (its statement, "and not from its Gaussianity", goes beyond its criterion: G1 keeps half of the excess of null points and, as Section 5.5 shows, overshoots the reference energy by 9 % at restart):

- the magnetic energy balance residual of G1 is 0.072, against 0.998 for G0;
- its error on the null count is 1.10, against 2.24 for G0 and 1.66 for W9;
- its restart drift of the magnetic energy at $\tau = 4$ is 0.064, against 0.21 for G0 and 0.12 for W9;
- and its energy-distance ratio on the held-out η is 2.55, against 3.5 for G0 and 9.3 for W9: G1 is the closest law of this report in that distance.

G1 uses the training values only: its samples at a held-out η are Gaussian draws from an interpolated covariance, not copies of snapshots. Its five seeds are five draws from one fitted law; the comparison was therefore repeated on the second realisation of the reference data, that of Section 5.9, with the same outcome on each point: 2.0 floors for G1 against 8.1 for W9 and 2.9 for G0, a magnetic balance residual of 0.072 (W9: 0.050), a null-count error of 1.00 against 1.56, a restart drift of the magnetic energy at $\tau = 4$ of 0.06 against 0.13 (`experiments/results/replications/e4_g1indep`).

What this establishes is limited and precise. The covariance between Fourier modes is enough to give a Gaussian law a magnetic production of the right order (0.93 to 1.28 of its Ohmic dissipation, where the mode-wise law has none) and half the spurious null points of the mode-wise law. The advantage of W9 and W11 over G0 on these metrics is therefore not evidence that they hold non-Gaussian structure, which is what the first version of this report claimed.

It does not establish that the learned models hold none, and G1 is not better than them everywhere:

- *Magnetic energy balance.* On the mean over the held-out η, W9 (0.055), the tilted ensemble W13 (0.038) and the widest plain network (0.037) are better than G1 (0.072), on both data realisations for W9; at their worst η, W9 and G1 are comparable. The production of G1 over its Ohmic dissipation is 0.93, 1.12 and 1.28 at the three held-out η, by decreasing η; that of W9 is 0.79, 0.93 and 1.02. A Gaussian law has no third-order cumulant, and at the smallest η G1 over-produces by 28 %. The lower residual of W9 is not an advantage, however: the split of the production in Section 5.5 shows that it results from two errors that cancel.
- *Restart test.* The drift at $\tau = 4$ ranks G1 (0.064) before W9 (0.124) for the magnetic energy and W9 (0.053) before G1 (0.095) for the kinetic energy. This ranking does not survive the anchoring on the reference energy: the drift of W9 is the recovery of a deficit of magnetic energy and that of G1 an excursion away from the right energy (Section 5.5, Figure 5.6). G0 is far behind on both readings.

- *Fluctuations of the energy budget.* The standard deviation of $\mathrm{d}E/\mathrm{d}t$ over that of the reference is 3.0 for G1, 1.6 for W9 and 1.4 for G0.
- *Null points.* Among the laws within 60 floors of the reference, G1 has the lowest error; several laws far from the reference have a lower one (the diffusion model, the rescaled ensemble, the widest plain network), for reasons that have nothing to do with a correct topology.

So: no learned model beats G1 in energy distance, and several are better than both Gaussian laws on the magnetic energy balance. Whether such advantages are non-Gaussian features that the models learned from the reference law could not be said from these runs; it is the question of the next section.

Table 5.1: Gaussian laws and mechanisms on the model with the affine term (measured; `experiments/results/replications/e4_mech`, five seeds, held-out η; the models G0, W9 and W11 are retrained in this job). G1: Gaussian law with the full covariance of the latent. W12: W9 fine-tuned with the stationarity loss, and its control. W13: exponential tilting of the second half of the control samples, to compare with the untilted half. Columns and notation as in Table 6.1.

Variant	Acts through	ED / floor, training η	ED / floor	div B	energy residual	magnetic balance	spectrum of B	magnetic helicity	null points	restart drift E_B	restart drift E_u	polarity	Seeds
G0 Gaussian per mode (no network)	family	2.67(8)	3.52(7)	1.5×10^{-8}	0.0015(3)	0.9981(9)	0.024(3)	0.035(5)	2.235(7)	0.212(3)	0.026(5)	0.018(3)	5
G1 Gaussian, full covariance (no network)	family	1.90(5)	2.55(4)	1.5×10^{-8}	0.0034(6)	0.0720(5)	0.0265(19)	0.037(4)	1.097(11)	0.0644(19)	0.095(3)	0.0148(19)	5
W9 mode-wise affine term	design	5.51(9)	9.3(3)	1.5×10^{-8}	0.0567(17)	0.0553(6)	0.145(3)	0.196(4)	1.665(14)	0.124(3)	0.053(6)	0.019(4)	5
W11 anchor + affine term	design	5.52(13)	8.6(2)	1.5×10^{-8}	0.0515(9)	0.0598(6)	0.143(4)	0.193(5)	1.696(12)	0.120(2)	0.053(3)	0.0169(10)	5
W12 control (same fine-tuning, no identity)	loss	5.46(5)	9.3(4)	1.5×10^{-8}	0.0567(18)	0.0573(8)	0.143(4)	0.195(6)	1.683(11)	0.123(4)	0.053(6)	0.019(2)	5
W12 affine term + stationarity loss	loss	9.5(5)	16.4(7)	1.5×10^{-8}	0.0292(11)	0.113(4)	0.218(7)	0.305(6)	1.981(20)	0.080(5)	0.139(6)	0.015(3)	5
W12 control, half ensemble	loss	4.59(11)	7.3(3)	1.5×10^{-8}	0.0564(19)	0.0564(5)	0.142(4)	0.192(7)	1.669(14)	–	–	0.030(5)	5
W13 affine term + exponential tilting	sampling	3.3(3)	4.6(3)	1.5×10^{-8}	0.011(2)	0.038(4)	0.081(8)	0.088(12)	1.25(2)	–	–	0.083(17)	5

5.5 Beyond a Gaussian law: three tests

The review of the third version of this report asked the question that the comparison with G1 left open, and showed that the restart drift read at one time could not answer it. Three hypotheses were registered (H25 to H27), their protocol attacked by the `jury-externe` role before the run, and two jobs of five seeds were run, one on each realisation of the reference data (`experiments/results/replications/e4_nongauss` and `e4_nongauss_indep`). G0, G1, W9 and W11 are retrained in these jobs and reproduce the earlier runs. Table 5.2 gives the measurements on the data of the main run; the second realisation gives the same verdicts.

The production split (H25). The mean magnetic production splits exactly into two parts,

$$\mathbb{E}\langle j \cdot (u \times B)\rangle \;=\; \mathbb{E}\langle j \cdot (\bar{u} \times B)\rangle \;+\; \mathbb{E}\langle j \cdot ((u-\bar{u}) \times B)\rangle, \qquad \bar{u} = \mathbb{E}[u].$$

The first is carried by the mean flow and the second-order statistics of B; the second is a third-order moment of the fluctuations and vanishes for any Gaussian law. In the reference (Figure 5.5), the third-order part is $+0.054 \pm 0.008$, -0.137 ± 0.004 and -0.270 ± 0.007 of the Ohmic dissipation at the three held-out η, by decreasing η (jackknife over the trajectories, the mean flow recomputed each time). Over the eleven values of η it decreases steadily from $+0.14$ at $\eta = 0.08$ to -0.31 at $\eta = 0.03$ and changes sign near $\eta = 0.066$. The reference law is measurably non-Gaussian in the very term that makes it a dynamo, and at the smallest held-out η the fluctuations remove about a fifth of what the mean flow produces. This is why G1, which has the mean-flow part right within 2 % and no third-order part (below 0.002 for the four Gaussian laws of the table), over-produces by 28 % there.

The hypothesis was that W9 holds none of it. It is **refuted**: the third-order part of W9 is -0.082 and -0.118 at the two lower η, with the sign of the reference in the five seeds, that is 0.60 and 0.44 of the reference value (0.63 and 0.41 on the second realisation); W11 gives the same figures. The Gaussian law fitted to the samples of W9 (G2, below) has none, as it must, which checks the estimate: this is structure of the generated law beyond its first two moments. It is partial: at the largest held-out η, where the third-order part of the reference is positive, that of W9 is small and of the opposite sign (-0.006), and the ratio to the reference is between 0.5 and 1.5 at one held-out value only. This is not an error of interpolation in η: at the three training values where the models are evaluated ($\eta = 0.08$, 0.054 and 0.03), the third-order part of W9 is $+0.008$, -0.069 and -0.121, that is 0.06, 0.73 and 0.39 of the reference. And it comes with a second-order error: the mean-flow part of W9 is 10 to 16 % below that of the reference at every held-out η. The magnetic energy balance of W9 (production over dissipation 0.79, 0.93 and 1.02) is therefore the sum of two errors: a mean-flow part too small by 0.15, 0.13 and 0.13 of the dissipation, and a third-order part off by -0.06, $+0.06$ and $+0.15$. At the two lower η they have opposite signs and cancel, in part at $\eta = 0.049$ (-0.07 is left) and almost exactly at $\eta = 0.033$ ($+0.02$); at the largest they add, and W9 under-produces by 21 %. Its lower residual on the mean over the held-out η (0.055 against 0.072 for G1) comes from this cancellation and is not an advantage.

The restart anchored on the reference (H26). With $a(\tau) = E_B(\tau)/E_{B,\mathrm{ref}} - 1$, reference snapshots stay within about 1 % of their stationary energy over 16 time units (Figure 5.6). W9 starts 12.3 % below it and comes back to it: -8.5 % at $\tau = 1$, -1.4 % at $\tau = 4$, -0.7 % at $\tau = 16$, with an overshoot of 0.0045 on the mean over the held-out η (0.014 at the largest). G1 starts at the right energy (-0.3 %), rises to $+10$ % at $\tau = 2$ and comes back to $+1.2$ % at $\tau = 16$ (overshoot 0.091). G0 starts at -2 %, falls to -18 % at $\tau = 0.5$, rises to $+19$ % at $\tau = 4$ and is still at $+3$ % at $\tau = 16$ (overshoot 0.173). H26 is **confirmed** on its registered criterion, each clause in every seed and on both realisations of the data.

The review of this result by the `encadrant` role adds what the criterion does not show. The overshoot gives each law an allowance equal to its own initial deviation, and the six laws of this job that start on the reference energy (within 0.04) or below it (G0, G1, G2, G3, W9, W11) all end up gaining magnetic energy, by 0.10 to 0.21 of the reference energy: the rise $\max_\tau a(\tau) - a(0)$ is 0.12 for W9, 0.11 for G1, 0.16 for G2 and 0.21 for G0, which first falls by 16 % (W10, which starts 21 % above the reference, relaxes downwards). W9 starts 12 % low, so its rise lands on the reference; G1 starts on the reference, and a rise of 0.11 takes it away. That the two rises have one cause is not established: W9 may simply relax to the stationary energy, as W10 does from above. The older runs stop at $\tau = 4$, where the six laws above have gained 0.06 to 0.21: there, with an approximate initial energy, the model fine-tuned with the stationarity loss and the network of width 2048 have gained 0.07 and are still rising (0.11 and 0.03 below the reference), which does not contradict the range; the network of width 5632, which starts 34 % low, has gained 0.29, above it. The largest distance to the reference energy over the whole course is comparable for the two (0.13 for W9, at the start; 0.10 for G1, at $\tau = 2$; 0.11 and 0.12 on the second realisation). The result therefore does not show that the states of W9 are closer to the attractor than those of G1. It shows that the restart drift of the objectives, relative to each ensemble's own initial energy, cannot be read as a ranking: its value for W9 (0.124 at $\tau = 4$) is almost entirely the recovery of the initial deficit, and its value for G1 (0.064) is an excursion on its way back. Neither law is stationary under the solver. The excursion of G1 is present at each held-out η, including the largest, where its production is below its dissipation: its instantaneous magnetic balance does not explain it, and no explanation was tested.

The Gaussian law with the moments of W9 (H27). G2 is the Gaussian law with the mean and the full covariance of 8192 samples of W9 at each η; the same construction applied to samples of G1 (G3) is the null control. The hypothesis was that G2 would overshoot more than W9 by at least 0.02, that is, that the course of W9 at restart needs its non-Gaussian part. The paired difference is -0.0051 ± 0.0005 on the data of the main run and -0.0056 ± 0.0005 on the second realisation, negative in every seed: G2 starts from the same deficit and shows no overshoot either. The null control passes on the second realisation (-0.0004 ± 0.0036 against a threshold of 0.02) and misses its threshold by 0.004 on the first (-0.024 ± 0.013; the review traced it to the sampling noise of the initial energy of the 128 restart samples, for laws that start near zero). By the letter of the criterion the test is invalid on the first realisation, and H27 is **refuted** on the second.

What this refutation allows is narrow. The registered measure does not separate W9 from the Gaussian law with its first two moments, because it cannot see an excursion beyond the reference smaller than the initial deficit; it does not show that the two behave alike. Computed after the fact, and therefore a **conjecture** until it is registered and tested, the excursion of G2 above the reference energy is larger than that of W9 by 0.024 ± 0.004 (0.026 ± 0.003 on the second realisation, in the five seeds of each): G2 rises faster and passes 2.4 % higher (2.6 % on the second realisation). A missing third-order sink of production would give this at the two lower η (production over dissipation 1.01 and 1.13 for G2, 0.93 and 1.02 for W9). The differences per η do not support that reading: the difference decreases as the missing third-order part grows (0.039, 0.021 and 0.011 for production gaps of 0.00, 0.08 and 0.12) and is not significant at the smallest η (1.5 standard errors, 3 of 5 seeds).

Comparing W9 with G2 isolates what its non-Gaussian part changes on the other metrics. It changes the third-order production (lost in G2), the fluctuation of the energy budget (the standard deviation of $\mathrm{d}E/\mathrm{d}t$ over that of the reference is 1.63 for W9 and 2.45 for G2) and the course of both energies at restart. It changes the null count little: the error of G2 is higher by 0.04 ± 0.03 (on the second realisation 0.04 ± 0.02), out of 1.7. And the energy distance barely sees it: samples of G2 and of W9 are 0.43 reference floors apart on the 21 features, about twice the

value expected for two samples of one law at this size and below the resolution of the objective (one floor), and G2 is 0.6 to 0.9 floors farther from the reference than W9.

So the answer to the question of this section has two halves. The model with the affine term is not a Gaussian law: it holds a third-order production of the sign of the reference and of 0.6 and 0.4 times its size at two held-out values out of three. But the first protocol could hardly show it. On the null count, one of the two metrics in which the model had seemed to stand out, its non-Gaussian part changes little (0.04 out of 1.7). On the other, the restart, the overshoot registered for the test does not separate the model from its Gaussianisation; the drift at $\tau = 4$ does (0.124 against 0.170 for G2, in the five seeds on both realisations), as does the excursion above the reference computed after the fact. The converse holds for the reference. The Gaussian law with the mean and covariance of the reference itself (G1) has 2.1 times its null points on the mean over η, at the training η as at the held-out ones (2.5 times at the largest η, 1.6 at the smallest), and at the held-out η, the only ones where the restart was run and where G1 is interpolated, it leaves the stationary energy by 9 to 11 %. The low null count of the reference is a non-Gaussian property, and so is its stationarity under the solver as far as G1 at interpolated η shows; no model of this report has learned either.

Table 5.2: Production split and restart anchored on the reference energy (measured; `experiments/results/replications/e4_nongauss`, five seeds; notation as in Table 6.1). Mean-flow and third-order parts of the mean magnetic production over the mean Ohmic dissipation, at the three held-out η. $a_B(0)$: initial deviation of the magnetic energy from the stationary energy of the reference; overshoot: $\max_\tau |a_B(\tau)| - |a_B(0)|$ over $\tau = 0.125, \dots, 16$; last column: $|a_B(16)|$; means over the held-out η. G2 and G3: Gaussian laws with the mean and covariance of 8192 samples of W9 and of G1. Reference row: jackknife standard errors for the production, and for the restart the mean curve of 4 draws of 128 snapshots.

Law	mean-flow part, $\eta =$ 0.0725	mean-flow part, $\eta =$ 0.0490	mean-flow part, $\eta =$ 0.0331	third-order part, $\eta =$ 0.0725	third-order part, $\eta =$ 0.0490	third-order part, $\eta =$ 0.0331	$a_B(0)$	overshoot of E_B	$\lvert a_B \rvert$ at $\tau = 16$
reference (SE: jackknife over trajectories; 4 restart draws)	0.946(8)	1.137(4)	1.270(7)	0.054(8)	-0.137(4)	-0.270(7)	0.00388	0.00108	0.00375
G0 Gaussian per mode (no network)	-0.0010(18)	-0.0022(5)	-0.002(3)	2.9×10^{-4}	2.3×10^{-4}	-0.0014(6)	-0.020(4)	0.173(3)	0.028(3)
G1 Gaussian, full covariance (no network)	0.9281(14)	1.1174(17)	1.2797(15)	-4.0×10^{-4}	-7.5×10^{-4}	0.0018(8)	-0.003(3)	0.091(8)	0.014(4)
G3 Gaussian law fitted to G1 samples (null control)	0.9235(12)	1.1243(15)	1.270(3)	2.2×10^{-4}	0.0013(7)	0.0017(8)	-0.011(4)	0.068(8)	0.0175(18)
W9 mode-wise affine term	0.7946(12)	1.007(3)	1.1368(19)	-0.0055(7)	-0.0817(10)	-0.1180(13)	-0.123(6)	0.0045(3)	0.0102(9)
G2 Gaussian law fitted to W9 samples	0.791(5)	1.010(4)	1.134(4)	-7.1×10^{-4}	-4.4×10^{-4}	-4.7×10^{-5}	-0.129(9)	-6.0×10^{-4}	0.016(3)
W11 anchor + affine term	0.7769(9)	1.008(2)	1.141(6)	-0.0096(3)	-0.0813(13)	-0.122(2)	-0.1220(14)	0.0060(2)	0.0096(20)
W10 anchored on the Gaussian field	0.448(3)	0.593(2)	0.670(3)	-0.0120(12)	-0.0426(15)	-0.0637(7)	0.211(3)	-0.0449(7)	0.0142(9)

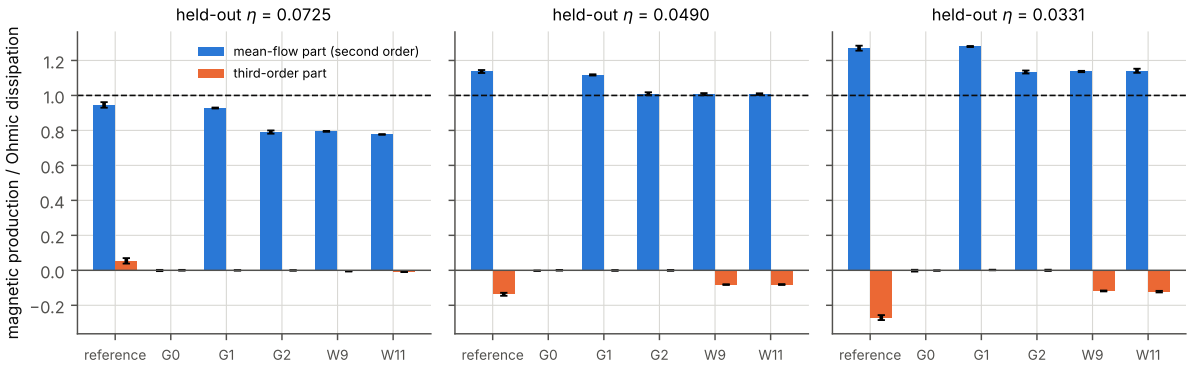


Figure 5.5: Mean magnetic production over the Ohmic dissipation, split into the part carried by the mean flow and the third-order part, at the three held-out η: reference (± 2 jackknife standard errors over the trajectories), the Gaussian laws G0, G1 and G2 and the learned models W9 and W11 (seed means, ± 2 standard errors over the seeds); the two parts of a stationary law sum to one (dashed line). Data plot from `experiments/results/replications/e4_nongauss`.

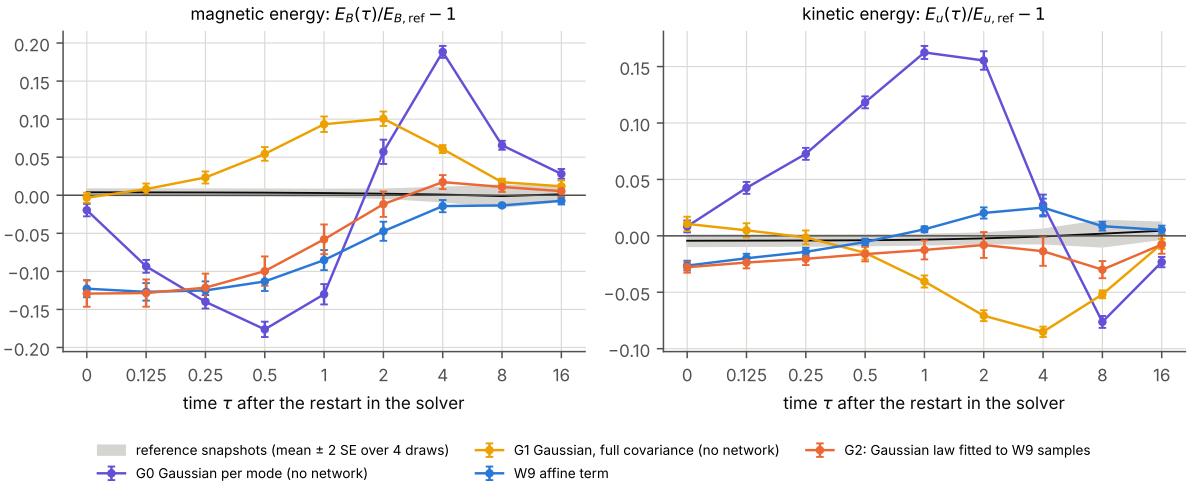


Figure 5.6: Restart test anchored on the reference: relative deviation of the ensemble-mean magnetic and kinetic energies from the stationary energies of the reference, when the solver restarts from generated snapshots, up to 16 time units; mean over the held-out η and the seeds (bars: ± 2 standard errors over the five seeds); grey band: reference snapshots, mean ± 2 standard errors over 4 draws of 128. Data plot from `experiments/results/replications/e4_nongauss`.

5.6 Loss and sampling mechanisms on a model close to the reference

The loss and sampling mechanisms of the first protocol were tried on a model two hundred floors from the reference, where they could only trade one error for another. Applied to the model with the affine term (Table 5.1), they behave differently from each other.

The stationarity loss still buys its target and now costs the law (H23). Fine-tuning W9 with the loss halves the residual of the total energy balance against the control (0.029 against 0.057, in every seed), and lowers the restart drift of the magnetic energy at $\tau = 4$ (0.080 against 0.123),

which is not an improvement: anchored on the reference energy, the fine-tuned model starts 17 % below it and is still 11 % below at $\tau = 4$, where the control goes from -12 % to -2 % (the smaller drift is a slower recovery from a larger deficit). The energy-distance ratio rises from 9.3 to 16.4, the residual of the magnetic energy balance doubles (0.113), the residual of the magnetic-helicity identity, the second target of the loss, gets worse (0.024 against 0.014), the error on the null count rises (1.98 against 1.68) and the drift of the kinetic energy nearly triples. The registered criterion allowed a degradation of 10 % of the distance: H23 is **refuted** on that clause. A penalty on two moments of the generated law, at the weight chosen for the first protocol, moves the law away from the data.

The tilting works when the law is close (H24). On the first-generation model the weights collapsed onto one sample. On the samples of the model with the affine term the effective sample size is 17 % of the sample on average (it is below 10 % in 5 of the 15 cases, with a minimum of 5 %), and H24 is **confirmed** as registered: the residual of the energy balance of the tilted ensemble is 0.011 against 0.056 for the untilted half, in every seed. The tilted ensemble is also better on the energy-distance ratio (4.6 against 7.3, and 3.3 once the bias of a weighted sample is estimated and removed), on the magnetic spectrum (0.08 against 0.14), the helicity (0.09 against 0.19), the null count (1.25 against 1.67) and the magnetic balance (0.038 against 0.056). It is worse on five quantities: the residual of the magnetic-helicity identity, one of its two targets (0.027 against 0.016), the kinetic spectrum (0.04 against 0.02), the kinetic energy, the fluctuation of the energy budget, and the polarity asymmetry, which a weighted sample of some forty effective members cannot keep at the sampling floor.

So the place of a constraint and the quality of the model interact: the same identities that could not help a wrong model improve most statistics of a nearly right one when they are imposed at sampling time, by reweighting, and hurt it when they are imposed through the loss at this weight. In both places the energy identity is the one that responds; the magnetic-helicity identity does not.

5.7 Longer trainings and wider networks

Would the models with an affine term overtake the Gaussian law with more training? The two best models were retrained with eight times the iterations (W9L, W11L; Table 5.3, Figure 5.7), in a separate job whose base models reproduce those of the main run bit for bit.

The answer is no, and the result is a negative one. The flow-matching loss on training samples goes down (from 1.18 to 1.08), but the generated law moves away from the reference: 18.8 and 19.5 floors on the held-out η against 9.3 and 8.6, and 12 against 5.5 on the training η themselves. The effect is present in every seed on the mean over the held-out η; it is largest at the smallest η and not separable at the largest one for W11L. The longer training improves the topology (error on the null count 1.4 against 1.7) and keeps the magnetic energy balance (residual 0.05), while the residual of the total energy balance nearly doubles and the restart drift of the kinetic energy rises; the restart drift of the magnetic energy is not separable from that of the base models. Why a lower training loss gives a worse law was not established; a decrease of the loss that sharpens the velocity field beyond what 50 integration steps resolve is one **conjecture**, memorisation of the 14 400 training snapshots another.

The hypothesis registered for this run (H18) was narrower: that with the affine term and a long training, flow matching holds what a Gaussian law cannot, a lower restart drift of the magnetic energy and a lower error on the null count than G0. Both paired differences are positive in the five seeds (0.070 ± 0.011 and 0.86 ± 0.03): H18 is **confirmed** as registered. The credit does not go to the long training, though: the models at the base budget already meet both clauses,

and the long training does not improve the drift. Nor does the first clause mean what the hypothesis says: the drift relative to the initial energy is lower for the learned models than for G0 partly because of their second moments (the Gaussian law fitted to the samples of W9 is at 0.170 against 0.212 for G0) and partly because of their non-Gaussian part (0.124 for W9); and a Gaussian law, G1, is lower still (0.064), so the clause does not measure what a Gaussian law cannot hold (Section 5.5).

Table 5.3: Long trainings (measured; `experiments/results/replications/e4_final_long`, five seeds, held-out η unless said otherwise): Gaussian baseline, models with the affine term or the anchor at the base budget and with 8 times the iterations. Columns and notation as in Table 6.1; FM loss: flow-matching loss on a training sample ($\pi/2 = 1.571$ for the exact field of independent standard Gaussians).

Variant	Acts through	ED / floor, training η	ED / floor	div B	energy residual	magnetic balance	spectrum of B	magnetic helicity	null points	restart drift E_B	restart drift E_u	polarity	FM loss	Seeds
G0 Gaussian per mode (no network)	family	2.67(8)	3.52(7)	1.5×10^{-8}	0.0015(3)	0.9981(9)	0.024(3)	0.035(5)	2.235(7)	0.212(3)	0.026(5)	0.018(3)	–	5
W9 mode-wise affine term	design	5.51(9)	9.3(3)	1.5×10^{-8}	0.0567(17)	0.0553(6)	0.145(3)	0.196(4)	1.665(14)	0.124(3)	0.053(6)	0.019(4)	1.1760(15)	5
W9L affine term, 8× iterations	design	11.91(15)	18.8(14)	1.5×10^{-8}	0.096(5)	0.049(4)	0.112(15)	0.12(2)	1.38(3)	0.142(10)	0.077(11)	0.0147(14)	1.075(2)	5
W10 anchored on the Gaussian field	design	36.37(10)	43.6(2)	1.5×10^{-8}	0.1458(9)	0.3092(17)	0.2081(19)	0.096(7)	2.464(16)	0.122(2)	0.092(4)	0.0135(9)	1.2727(18)	5
W11 anchor + affine term	design	5.52(13)	8.6(2)	1.5×10^{-8}	0.0515(9)	0.0598(6)	0.143(4)	0.193(5)	1.696(12)	0.120(2)	0.053(3)	0.0169(10)	1.1742(14)	5
W11L anchor + affine, 8× iterations	design	12.0(5)	19.5(15)	1.5×10^{-8}	0.098(8)	0.043(3)	0.128(14)	0.14(2)	1.41(3)	0.154(17)	0.081(13)	0.018(5)	1.075(2)	5

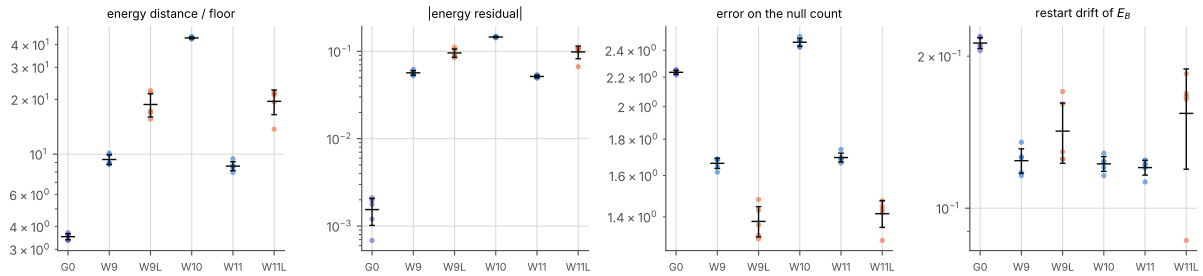


Figure 5.7: Gaussian baseline (violet), flow matching with the explicit affine term or the Gaussian anchor at the base budget (blue) and with 8 times the iterations (orange), on the held-out η: one dot per seed, seed mean ± 2 standard errors. Data plot from `experiments/results/replications/e4_final_long/`.

Is the failure of the plain networks a matter of width, as the rank argument of Section 3.5 suggests (H21)? The equivariant network with the smooth embedding and no affine term was trained at three widths, 512, 2048 and 5632 (the last one above the latent dimension of 5320), and at 8 times the iterations for the first two (Table 5.4, Figure 5.8). The five seeds come from a second scan, split over three GPU jobs whose reference data are identical; a first scan on a single GPU was cut by its time limit and agrees with it on the seeds it has.

- **Width 512** is the matched control of W9, the same network without the affine term: 205 floors, with a final loss of 1.61, above $\pi/2$. The affine term is worth a factor of 22.
- **Width 2048** is at 25.6 floors, with a loss of 1.22: eight times closer. Capacity matters. This network is farther from the reference than W9 in energy distance but better on the total energy residual (0.012 against 0.057); its restart drift of the magnetic energy at $\tau = 4$ is lower (0.072 against 0.124), which is not an improvement: anchored on the reference energy, it starts 10 % below and is 3 % below at $\tau = 4$, a slower recovery than W9 from a smaller deficit.
- **Width 5632**, above the latent dimension, reaches a much lower loss (0.78) and a worse law: 104 floors. The order registered for H21 (the wider the closer, and the widest at less than a fifth of the narrowest) does not hold, in any seed: H21 is **refuted**. This network is not uniformly worse, though: its null-count error (0.96) is lower than those of W9 (1.66) and of width 2048 (2.01), its magnetic balance residual (0.037) is the lowest of the networks, and its restart drift of the magnetic energy (0.45) the largest.
- **More training does not help.** At width 512 the loss falls to 1.52, now below $\pi/2$, and the law is slightly worse (215 floors, $+10 \pm 1.6$); at width 2048 the loss falls to 0.95 and the distance more than doubles (59 floors against 25.6), the damage being concentrated at the smallest η.

So the width is part of the story and not the whole of it. For these plain networks the training loss and the distance of the generated law to the reference part ways: beyond width 2048 at the base budget, a lower loss on the training samples comes with a worse law, at the training η as well as between them. With 14 400 training snapshots in 5320 dimensions, fitting the training set more closely than the affine model does (its loss is 1.18) is not learning the law; that this is memorisation of the snapshots is a **conjecture**, not a measurement. In energy distance, the explicit affine term at width 512 does better than any width tried; on other metrics the ranking of the networks is not the same.

Table 5.4: Width scan (measured; `experiments/results/replications/e4_scan2`, five seeds on three GPU jobs with the same data, held-out η): the equivariant potentials model with the smooth embedding and no affine term, for three widths and two training budgets, against the model with the affine term and the mode-wise Gaussian law. Columns and notation as in Table 6.1; FM loss as in Table 5.3. In these result files the field `cond_scale` of the configuration is the default of the run (8); the builders of these variants and of W9 fix the embedding of the condition to 0.5.

Variant	Acts through	ED / floor, training η	ED / floor	div B	energy residual	magnetic balance	spectrum of B	magnetic helicity	null points	restart drift E_B	restart drift E_u	polarity	FM loss	Seeds
S width 512	family	165.8(4)	205.3(5)	1.4×10^{-8}	0.2737(9)	0.3284(19)	0.663(5)	0.184(6)	2.116(11)	0.3725(16)	0.268(3)	0.013(2)	1.613(3)	5
S width 2048	family	25.8(2)	25.6(4)	1.5×10^{-8}	0.0117(8)	0.0951(3)	0.151(2)	0.286(4)	2.007(19)	0.0719(14)	0.075(4)	0.011(2)	1.222(2)	5
S width 5632	family	70(5)	104(8)	1.5×10^{-8}	0.274(16)	0.037(5)	0.34(3)	0.37(3)	0.96(5)	0.45(4)	0.27(3)	0.022(4)	0.777(3)	5
S width 512, 8× iterations	family	173.4(11)	215.5(19)	1.4×10^{-8}	0.260(2)	0.400(5)	0.563(10)	0.349(14)	2.202(17)	0.366(5)	0.223(5)	0.010(2)	1.5247(14)	5
S width 2048, 8× iterations	family	110(11)	59(3)	1.5×10^{-8}	0.150(7)	0.138(6)	0.291(11)	0.441(12)	2.14(3)	0.264(10)	0.30(2)	0.026(3)	0.949(3)	5
W9 mode-wise affine term	design	5.47(12)	9.3(3)	1.5×10^{-8}	0.0566(17)	0.0551(4)	0.144(3)	0.195(4)	1.669(16)	0.124(3)	0.052(6)	0.016(3)	1.1773(10)	5
G0 Gaussian per mode (no network)	family	2.67(8)	3.52(7)	1.5×10^{-8}	0.0015(3)	0.9981(9)	0.024(3)	0.035(5)	2.235(7)	0.212(3)	0.026(5)	0.018(3)	–	5

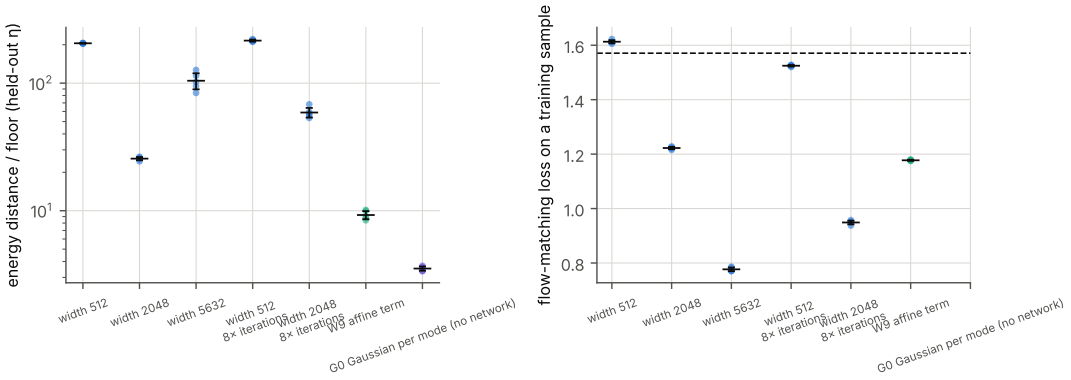


Figure 5.8: Width scan (hypothesis H21): energy-distance ratio and flow-matching loss for three widths and two budgets, against the model with the explicit affine term and the Gaussian baseline. Dashed line: $\pi/2$. Data plot from `experiments/results/replications/e4_scan2/`.

5.8 Topology: null points and the small scales

The number of magnetic null points is the most demanding of the metrics: no generator has it right. The penalty model and the diffusion model have too few (0.3 and 0.6 times the reference count); all the others have too many. A conjecture of the `physicien` role (H13) was that the count scales as the cube of the magnetic Taylor wavenumber k_λ, as it would for a Gaussian random field, so that the error on the count would be three times the relative error on k_λ. The fit over all generators, held-out values and seeds gives a slope close to five and not three (Figure 5.9): H13 was **refuted** on the slope. The correlation is strong, but it does not explain the excess: the fitted line does not pass through the origin, and a generator with exactly the reference k_λ would still have about 2.8 times the reference number of null points. The excess of nulls is a property of the generated fields beyond their small-scale energy.

5.9 Replication with other data

The five-seed runs made on pods of the same type reproduce one another to the last digits, because the data and model seeds are the same: they check the reproducibility of the pipeline, not the robustness of the conclusions. For that, the whole protocol (groups A to E, five seeds) was run once more on reference data regenerated with other seeds of the initial conditions (Table 5.5). The statistics of the two reference sets agree within their standard errors. The ratios of the replication differ from those of the main run by -18 % to $+12$ %; part of this comes from the noise floors (they differ by a factor of 1.6 at one held-out value) and part from the distances themselves, which are 8 to 13 % lower for the three best laws.

Every conclusion of this section is unchanged in the replication, and so is every verdict recomputed on it by the `encadrant`: the first-generation models are between 110 and 490 floors from the reference, in the same order (the only inversions are among the half-sample and tilted ensembles); the affine term brings the equivariant model from 191 to 8.2 floors (a factor of 23); the mode-wise Gaussian law is closer (2.9) and has no magnetic production; the anchored model without correction stays at 43 floors; the null-count errors and restart drifts keep their order.

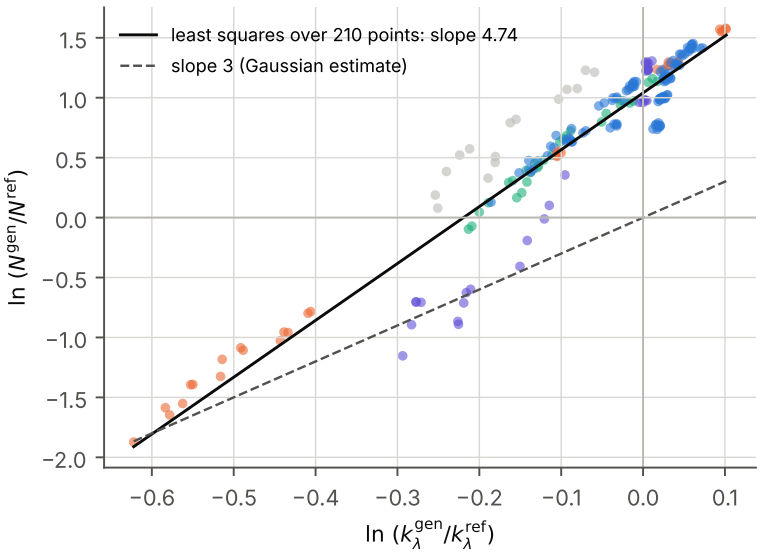


Figure 5.9: Number of magnetic null points against the magnetic Taylor wavenumber, generated over reference, one point per variant, held-out η and seed; the fit is the one of the digest (hypothesis H13). Data plot from `experiments/results/e4_mhd3d_s<k>.json`.

Table 5.5: Main run ("main") against a replication in which the reference data were regenerated with other seeds of the initial conditions ("other": `E4_DATA_SEED=100`; `experiments/results/replications/e4_indep`); five seeds each, held-out η, notation of Table 6.1.

Variant	ED / floor, main	ED / floor, other	magnetic balance, main	magnetic balance, other	null points, main	null points, other	restart drift E_B, main	restart drift E_B, other
W0 free components	236(6)	221(8)	0.159(15)	0.162(12)	1.11(5)	0.98(7)	0.513(9)	0.520(10)
W1 divergence penalty	464(12)	420(7)	0.410(5)	0.395(6)	0.698(12)	0.686(7)	0.5008(14)	0.4995(20)
W2 final projection	179(8)	169(9)	0.159(15)	0.162(12)	0.91(5)	0.79(8)	0.443(14)	0.453(16)
W3 projected flow	177(7)	164(8)	0.161(12)	0.165(13)	0.90(7)	0.82(8)	0.448(17)	0.452(19)
W4 vector potentials	256(7)	237(6)	0.149(14)	0.153(18)	1.14(7)	1.05(8)	0.517(10)	0.517(10)
W4s potentials, B → -B equivariant	209.8(19)	191(3)	0.264(14)	0.256(12)	1.83(7)	1.68(6)	0.382(12)	0.385(12)
W5 stationarity loss	215.0(19)	196.7(8)	0.516(13)	0.516(8)	2.36(2)	2.209(17)	0.334(3)	0.334(2)
W5 control (same fine-tuning, no identity)	210(2)	192(3)	0.251(15)	0.244(12)	1.79(7)	1.64(6)	0.390(12)	0.392(11)
W6 diffusion (VP, v-prediction)	519(17)	491(10)	0.213(11)	0.212(12)	0.43(2)	0.456(19)	0.637(2)	0.632(2)
W7 exponential tilting	195(11)	214(6)	0.43(4)	0.41(4)	2.24(19)	2.05(6)	–	–
W8 per-sample balance rescaling	117(2)	110(3)	0.195(19)	0.191(17)	0.71(8)	0.63(7)	–	–
W4c potentials, smooth condition	215.4(3)	205.8(7)	0.3843(16)	0.387(3)	2.142(12)	2.018(12)	0.3826(11)	0.3807(16)
G0 Gaussian per mode (no network)	3.52(7)	2.90(5)	0.9981(9)	0.9984(8)	2.235(7)	2.102(7)	0.212(3)	0.210(3)
W9 mode-wise affine term	9.3(3)	8.20(9)	0.0553(6)	0.0506(6)	1.665(14)	1.551(8)	0.124(3)	0.122(4)
W10 anchored on the Gaussian field	43.6(2)	43.3(2)	0.3092(17)	0.3122(19)	2.464(16)	2.323(15)	0.122(2)	0.125(4)

Variant	ED / floor, main	ED / floor, other	magnetic balance, main	magnetic balance, other	null points, main	null points, other	restart drift E_B, main	restart drift E_B, other
W11 anchor + affine term	8.6(2)	7.28(13)	0.0598(6)	0.0547(7)	1.696(12)	1.565(8)	0.120(2)	0.121(3)

5.10 Objectives of the race

The objectives of the team were quantified on the first day, before any model was trained (`race/teams/claude/objectives.toml`), and are measured by `experiments/race_metrics.py` on one generator, chosen on the validation values (Table 5.6).

Generator measured by C1–C5, T1 and T2: `W9_skip_sym`, the learned variant of the main run with the lowest energy-distance ratio on the validation η (file `experiments/results/e4_selection.json`).

Table 5.6: Objectives of the race against the measured values (generated by running the command of each objective).

Objective	Target	Measured	Seeds	Met
C1-3d-generator	≤ 10	9.35	5	yes
C2-divergence	$\leq 1.0 \times 10^{-6}$	1.6×10^{-8}	5	yes
C3-energy-balance	≤ 0.05	0.0567	5	no
C4-spectra-helicity	≤ 0.15	0.341	5	no
C5-topology	≤ 0.2	1.66	5	no
C6-comparative-study	≥ 6	20	5	yes
C7-toolbox	≥ 6	15	0	yes
T1-restart-drift	≤ 0.05	0.124	5	no
T2-polarity-symmetry	≤ 0.03	0.0186	5	yes

Five of the nine objectives are met: the distance to the reference law (C1, below ten floors), the divergence (C2), the number of decided comparisons (C6), the toolbox (C7) and the polarity symmetry (T2) reach their targets; the energy balance (C3) misses its target narrowly; the spectra and helicity (C4), the topology (C5) and the restart drift (T1) miss theirs by factors of two to eight. The targets were set before any model existed and were not changed. Three caveats on this table. The selection between W9 and W11 on the validation values is inside the seed noise (paired difference -0.29 ± 0.60): the measured generator could as well have been W11, with the same outcome for every objective. The objective C1 is met inside the uncertainty of the floor: the ratio is 9.35 with the floor of the protocol and 9.96 with the floor averaged over 20 splits, and one seed of the five is above 10 with the first floor, two with the second. And the selected generator is the best *learned* one: the Gaussian baselines, which have lower ratios on the validation values (2.5 for G1 and 4.0 for G0, against 8.9 for W9), are not candidates, because the objective written on the first day names the model family (conditional flow matching on Fourier modes); the line of code that excludes them was written after the results of G0 were known. G0

would meet seven of the nine objectives, missing the topology and the restart drift. A fourth caveat concerns T1 itself: the drift relative to the ensemble's own initial energy, read at one time, is a poor measure of what it was written for. Its value for W9 (0.124) is almost entirely the recovery of a 12 % deficit of magnetic energy, and a law with the right energy and the same rise would have the same drift, away from the reference instead of towards it: the number does not say which (Section 5.5).

6 Comparative study

Table 6.1 is the comparison the offer asks for: one line per mechanism or model family, one column per metric, on the three held-out values of η, seed mean and standard error. Table 6.2 gives, for each registered hypothesis, the status after review. Both are generated from the result files and the register; the sortable version of the first is in the explorer.

Table 6.1: Comparative table of E4 (measured; `experiments/results/e4_mhd3d_s<k>.json`). ED / floor: energy distance to the reference over the noise floor, at the training and at the held-out η; div B: largest relative divergence of B; energy residual: absolute residual of the mean energy balance over the mean dissipation; magnetic balance: absolute residual of the mean magnetic energy balance (0: production equals Ohmic dissipation; 1: no production); spectrum of B: relative L^2 error; magnetic helicity and null points: relative errors of the means; restart drift: relative drift of the magnetic and of the kinetic energy after $\tau = 4$; polarity: asymmetry of the sign of the cross helicity. Lower is better in every column except the two restart drifts, which do not rank the laws (Section 5.5). Notation: 217(14) is the seed mean 217 with a standard error of 14 on its last digits; values below 10^{-3} are given without error. The tilted ensemble W7 is a weighted sample whose effective size is about one sample: its line describes that sample. No restart test for W7 and for the rescaled ensemble W8.

Variant	Acts through	ED / floor, training η	ED / floor	div B	energy residual	magnetic balance	spectrum of B	magnetic helicity	null points	restart drift E_B	restart drift E_u	polarity	Seeds
W0 free components	none	165.4(5)	236(6)	0.454(2)	0.314(11)	0.159(15)	2.37(13)	1.99(19)	1.11(5)	0.513(9)	0.413(9)	0.015(3)	5
W1 divergence penalty	loss	275(4)	464(12)	0.261(6)	0.613(18)	0.410(5)	23.0(17)	38(3)	0.698(12)	0.5008(14)	0.709(4)	0.020(4)	5
W2 final projection	sampling	106.9(6)	179(8)	3.1×10^{-8}	0.314(11)	0.159(15)	2.04(13)	1.99(19)	0.91(5)	0.443(14)	0.354(11)	0.015(4)	5
W3 projected flow	sampling	105.8(6)	177(7)	2.6×10^{-8}	0.312(16)	0.161(12)	2.00(18)	1.9(2)	0.90(7)	0.448(17)	0.355(16)	0.015(2)	5
W4 vector potentials	design	175.4(5)	256(7)	1.4×10^{-8}	0.408(12)	0.149(14)	2.45(19)	2.1(3)	1.14(7)	0.517(10)	0.425(12)	0.013(3)	5
W4s potentials, B $\to$ -B equivariant	design	166.3(3)	209.8(19)	1.4×10^{-8}	0.284(10)	0.264(14)	0.97(8)	0.45(9)	1.83(7)	0.382(12)	0.281(16)	0.013(3)	5
W5 stationarity loss	loss	165.1(5)	215.0(19)	1.4×10^{-8}	0.191(5)	0.516(13)	0.473(15)	0.693(9)	2.36(2)	0.334(3)	0.243(3)	0.013(2)	5
W5 control (same fine-tuning, no identity)	loss	166.4(3)	210(2)	1.4×10^{-8}	0.290(10)	0.251(15)	1.03(9)	0.50(11)	1.79(7)	0.390(12)	0.289(16)	0.011(2)	5
W6 diffusion (VP, v-prediction)	family	160.4(6)	519(17)	1.3×10^{-8}	0.747(14)	0.213(11)	15.8(16)	22(2)	0.43(2)	0.637(2)	0.616(14)	0.022(6)	5
W7 exponential tilting	sampling	161(9)	195(11)	1.4×10^{-8}	0.204(8)	0.43(4)	0.51(6)	0.41(11)	2.24(19)	–	–	0.47(3)	5
W8 per-sample balance rescaling	sampling	92.9(3)	117(2)	1.4×10^{-8}	$< 10^{-12}$	0.195(19)	0.86(8)	0.52(11)	0.71(8)	–	–	0.0134(18)	5
W4c potentials, smooth condition	design	175.1(8)	215.4(3)	1.4×10^{-8}	0.2694(6)	0.3843(16)	0.637(6)	0.330(5)	2.142(12)	0.3826(11)	0.252(4)	0.0139(19)	5
G0 Gaussian per mode (no network)	family	2.67(8)	3.52(7)	1.5×10^{-8}	0.0015(3)	0.9981(9)	0.024(3)	0.035(5)	2.235(7)	0.212(3)	0.026(5)	0.018(3)	5
W9 mode-wise affine term	design	5.51(9)	9.3(3)	1.5×10^{-8}	0.0567(17)	0.0553(6)	0.145(3)	0.196(4)	1.665(14)	0.124(3)	0.053(6)	0.019(4)	5
W10 anchored on the Gaussian field	design	36.37(10)	43.6(2)	1.5×10^{-8}	0.1458(9)	0.3092(17)	0.2081(19)	0.096(7)	2.464(16)	0.122(2)	0.092(4)	0.0135(9)	5

W11 anchor + affine term	design	5.52(13)	8.6(2)	1.5×10^{-8}	0.0515(9)	0.0598(6)	0.143(4)	0.193(5)	1.696(12)	0.120(2)	0.053(3)	0.0169(10)	5

Table 6.2: Hypotheses of E4, registered with a refutation criterion before the computation (`research/hypotheses.json`), with their status after the review of the `encadrant` role.

Id	Hypothesis (shortened)	Status	Verdict of the `encadrant` role (beginning)
H4	In E4 (3D), generating vector potentials (W4_potentials, constraint by model design) gives a lower energy-distance ratio on the held-out η than generating free components (W0_free).	refuted	REFUTED (encadrant, 2026-11-12): W0_free - W4_potentials = -20.05 ± 3.21 (-6.2 SE), W4 below W0 in 0 of 5 seeds, negative at each held-out η; the two CPU seeds agree (-26.9).
H5	In E4, the divergence penalty on the end-of-path prediction (W1_penalty, weight 10; constraint by a training loss) does not enforce the constraint: […]	supported	CONFIRMED (encadrant, 2026-11-12): relative divergence of B 0.206 against 0.380 (ratio 0.544, not below one tenth); W0_free - W1_penalty = -228.2 ± 12.7, 0 of 5: the penalty does not enforce the constraint and degrades the law.
H6	In E4, the projected flow (W3_projected_flow: projected noise and projection of every integration stage) gives a lower energy-distance ratio on the held-out η than the Leray projection of the final samples of the free model (W2_final_projection).	refuted	REFUTED (encadrant, 2026-11-12): W2_final_projection - W3_projected_flow = 2.69 ± 3.72 (0.72 SE), 4 of 5 seeds: no difference at 2 SE; the CPU seeds show the same absence of effect.
H7	In E4, fine-tuning the equivariant potentials model with the law-level stationarity loss (W5_stationarity: energy identity and magnetic-helicity identity $E[\langle j \cdot B \rangle] = 0$, imposed at every η of the range) lowers the absolute residual of the mean energy balance on […]	supported	CONFIRMED (encadrant, 2026-11-12): |energy residual| W5_control - W5_stationarity = 0.0988 ± 0.0108 (9.1 SE), 5 of 5 seeds (0.290 → 0.191); ED ratio W5/control = 1.023.
H8	In E4, imposing the stationarity identities at training time (W5_stationarity) gives a lower energy-distance ratio on the held-out η than imposing the same identities at sampling time by exponential tilting of the samples of the control model (W7_tilting: […]	dropped	NOT DISCRIMINATING (encadrant, 2026-11-12): the registered criterion is formally met (W7_tilting - W5_stationarity_half = 30.0 ± 10.1, 5 of 5), but ESS/N is 1/256 in 14 of 15 cells: the tilted law is one sample and its energy distance is not a measurement of a law; with an estimate of the extra bias of a weighted sample the sign flips (-13.9 ± 9.5). […]
H9	In E4, the B → -B equivariant velocity field (W4s_potentials_sym, constraint by model design) lowers the polarity asymmetry $\lvert P(\langle u \cdot B \rangle > 0) - 1/2 \rvert$ of the generated law on the held-out η compared with the same model without symmetrization (W4_potentials), without […]	refuted	REFUTED (encadrant, 2026-11-12) on its first clause: asymmetry W4 - W4s = 0.0004 ± 0.0058, 3 of 5 seeds, both at or below the binomial floor 0.0176: the test could not discriminate at n = 512. The second clause holds: ED ratio W4s/W4 = 0.819 (-46.5 ± 4.8, 0 of 5).
H10	In E4, conditional flow matching (W4_potentials: linear path, Heun, 100 network evaluations) gives a lower energy-distance ratio on the held-out η than a variance-preserving diffusion with ancestral sampling (W6_diffusion: […]	supported	CONFIRMED as registered (encadrant, 2026-11-12): W6_diffusion - W4_potentials = 262.4 ± 12.1, 5 of 5, positive at each held-out η. Caveat of the encadrant: at the training η the diffusion model is better than flow matching (-15.06 ± 0.90, 0 of 5): the result is about interpolation in η with the aliased condition embedding, not about the model family at fixed η.

Id	Hypothesis (shortened)	Status	Verdict of the `encadrant` role (beginning)
H11	In the E4 reference data, the magnetic helicity spectrum is bipolar at every η: H(k = 1) < 0 and Σ_{k ≥ 2} H(k) > 0, with a large-scale energy fraction E_B(k = 1)/E_B between 0.2 and 0.8, so that the stationarity identity E[⟨j·B⟩] = Σ k² H(k) = 0 is a […]	refuted	REFUTED (encadrant, 2026-11-12): recomputed from the raw snapshots at the 11 η, independently of the experiment code, every value equal to e4_reference.json. H(1) < 0 at 11 of 11 η (z from -19 to -58) and E_B(1)/E_B in [0.370, 0.438] hold; Σ_{k ≥ 2} H(k) > 0 by 2 SE fails at η = 0.0365 and 0.03309 (z = -1.4 and -1.2, sign undetermined) and at η = 0.03 (z = -5.5, negative). […]
H12	In E4, the magnetic-helicity identity E[⟨j·B⟩] = 0 constrains the generated magnetic law more than the energy identity: on the samples of W4s_potentials_sym at the held-out η its normalized residual is at least twice that of the energy identity, and tilting […]	refuted	REFUTED (encadrant, 2026-11-12): seed mean of r_b/r_a = 0.504, r_b > r_a in 0 of 5 seeds. The tilting clause is met (|err H_M| tilt a - tilt b = 0.238 ± 0.095, 5 of 5) against a degenerate comparator (tilt on the energy identity: ESS/N 0.0096); tilt b against no tilt gains 0.110 ± 0.049; the two CPU seeds do not agree in sign.
H13	In E4, the error of a generator on the number of magnetic null points follows its error on the magnetic Taylor wavenumber k_λ = (⟨|j|²⟩/⟨|B|²⟩)^{}1/2 as N ∝ k_λ^3: […]	refuted	REFUTED on the slope (encadrant, 2026-11-12): slope 4.875, R^2 0.902 over 135 points (seed-bootstrap interval [4.83, 4.92]); CPU 4.96. Protocol check passed (reference nulls change by 0.6 to 3.6 % between 48^3 and 64^3).
H14	In E4, the relative divergence of the free model W0_free is a representation error of the network and not an integration error (Proposition 2, consequence (c)): […]	supported	CONFIRMED (encadrant, 2026-11-12): relative divergence of B of W0_free 0.3792 at 50 Heun steps and 0.3788 at 200, ratio 0.998 to 1.000 per seed, above 0.1: integration error is excluded; 'representation error' is the remaining reading, not a separate measurement.
H15	In E4, the error of the generators at the held-out η comes from the embedding of the condition: the Fourier embedding of the pre-race code has frequencies up to 8 cycles over the normalized parameter range, above the Nyquist frequency of the 6 training values […]	refuted	REFUTED (encadrant, 2026-11-13): ratio of the seed-mean ED ratios W4c/W4 = 215.4/256.3 = 0.84 on the held-out η, above the 0.5 of the criterion; the paired difference ED(W4) - ED(W4c) = 40.9 ± 6.8 (6.0 SE, 5 of 5 seeds) is real, and absent at the training η (0.34 ± 0.32): the embedding of the condition explains part of the interpolation error, not the failure.
H16	In E4, a conditional Gaussian model with one mean and one covariance per Fourier mode (G0_gaussian_modes: no network; moments estimated at the training η and interpolated linearly in log η, symmetrized under B → -B) has a lower energy-distance ratio on the […]	supported	CONFIRMED (encadrant, 2026-11-13): G0 3.52 against 177 to 519 for W0–W6 on the held-out η (W3 the lowest, 176.6); best of them per seed 153–194 against G0 3.36–3.70, 5 of 5 seeds.
H17	In E4, the over-dispersion of the flow-matching generators comes from the bottleneck of the network (width 512 for 5320 latent dimensions), which cannot represent the linear part of the velocity field: […]	supported	CONFIRMED (encadrant, 2026-11-13): ratio W9/W4s = 9.35/209.8 = 0.045, below 0.2; ED(W4s) - ED(W9) = 200.4 ± 2.2 (91 SE), 5 of 5 seeds; flow-matching loss of W9 on a training sample 1.18, below π/2. The factor is 22.4 at the held-out η and 30.2 at the training η.
H18	In E4, flow matching adds to the second-order baseline what a Gaussian law cannot hold: with the mode-wise affine term and a longer training (W9L_skip_long, 8 times the iterations of W9), the drift of the magnetic energy when the solver is restarted from the […]	supported	CONFIRMED as registered (encadrant, 2026-11-13): G0 - W9L on the held-out η, restart drift of E_B 0.070 ± 0.011 (6.4 SE, 5 of 5 seeds), null-count error 0.856 ± 0.035 (25 SE, 5 of 5). Caveats: W9 at the base budget already meets both clauses; […]

Id	Hypothesis (shortened)	Status	Verdict of the `encadrant` role (beginning)
H19	In E4, a flow-matching field anchored on the exact field of the mode-wise Gaussian law, v = v_Gauss(x, t, η) + network (W10_anchored: […]	refuted	REFUTED (encadrant, 2026-11-13): ED(W10) - ED(G0) = +40.0 ± 0.26 on the held-out η (first clause not met); restart drift of E_B: G0 - W10 = 0.090 ± 0.003, 5 of 5 (met); null-count error: G0 - W10 = -0.229 ± 0.014, 0 of 5 (not met).
H20	In E4, the anchored model fails (W10_anchored on CPU seeds 0 and 1: 45 times the floor against 3.3 for G0) because the network's own linear component perturbs the second-order statistics that the fixed Gaussian field had right. […]	refuted	REFUTED (encadrant, 2026-11-13) on its first clause: ED(W11) - ED(G0) = +5.08 ± 0.22 (24 SE), 0 of 5 seeds; the two other clauses are met: restart drift of E_B G0 - W11 = 0.092 ± 0.003 (5 of 5), null-count error G0 - W11 = 0.539 ± 0.014 (5 of 5). Reported with it: […]
H21	In E4, the over-dispersion of the flow-matching generators without an explicit linear term comes from the width of the network and not from the training budget: […]	refuted	REFUTED as registered (encadrant, 2026-11-13) on its first two clauses, 5 seeds, held-out η: S_w512 205.3 ± 0.5, S_w2048 25.6 ± 0.4, S_w5632 104.4 ± 7.5 floors: the order is not monotonic (2048 - 5632 = -78.8 ± 7.6, 0 of 5 seeds) and S_w5632 is at 0.51 of S_w512, above one fifth. Third clause not triggered: S_w512_long 215.5 ± 1.9 (1.05 of S_w512). […]
H22	In E4, the failures of the mode-wise Gaussian law G0 on the magnetic energy balance, on the restart drift of the magnetic energy and on the number of null points come from the independence of its Fourier modes and not from its Gaussianity: […]	supported	CONFIRMED as registered (encadrant, 2026-11-13): |magnetic balance| of G1_gaussian_full 0.072 (per seed 0.071–0.074), below 0.2; restart drift of E_B at $\tau = 4$, W9 - G1 = 0.0598 ± 0.0034 (17.6 SE, 5 of 5 seeds); null-count error, W9 - G1 = 0.568 ± 0.020 (29 SE, 5 of 5). Reported with it: ED ratio 2.55 against 9.35 (W9) and 3.52 (G0); […]
H23	In E4, in the regime where the generator is close to the reference law (the model with the affine term, W9_skip_sym), fine-tuning with the law-level stationarity loss on the energy and magnetic-helicity identities (W12_skip_stationarity: […]	refuted	REFUTED on its second clause (encadrant, 2026-11-13): the first clause is met, |energy residual| control - stationarity = 0.0275 ± 0.0012 (22 SE, 5 of 5 seeds); the energy-distance ratio of W12_skip_stationarity is 16.37 against 9.31 for the control, a factor 1.76 where the criterion allowed 1.10. Reported with it: […]
H24	In E4, exponential tilting on the energy and magnetic-helicity identities does not collapse when the generated law is close to the reference: […]	supported	CONFIRMED as registered (encadrant, 2026-11-13): ESS/N of W13_skip_tilting 0.172 on the held-out η (below 0.1 in 5 of the 15 cells, minimum 0.049); |energy residual| untilted half - tilted = 0.0449 ± 0.0030 (14.8 SE, 5 of 5 seeds). Reported with it: the tilted ensemble is better on the ED ratio (4.58 against 7.26; […]
H25	In E4, the flow-matching model with the affine term (W9_skip_sym) does not hold the third-order part of the mean magnetic production: […]	refuted	REFUTED (encadrant, 2026-11-13), on two realisations of the data: the third-order part of the production of W9 over D is -0.0055 ± 0.0007, -0.0817 ± 0.0010 and -0.1180 ± 0.0013 at η = 0.0725, 0.0490 and 0.0331 (same sign in 5 of 5 seeds; replication -0.0116, -0.0848, -0.1126), against +0.0542 ± 0.0078, -0.1371 ± 0.0043 and -0.2703 ± 0.0071 for the reference (jackknife): […]

Id	Hypothesis (shortened)	Status	Verdict of the `encadrant` role (beginning)
H26	In E4, anchored on the stationary energy of the reference, the restart of the model with the affine term is a monotone relaxation and that of the Gaussian laws is not: […]	supported	CONFIRMED on the registered criterion (encadrant, 2026-11-13), each clause in 5 of 5 seeds on two realisations of the data: overshoot of W9 0.0045 (replication 0.0048), |a(16)| of W9 0.0102 (0.0133), overshoot of G1 0.0914 ± 0.0078 (0.1029 ± 0.0026), of G0 0.1732 (0.1986). Caveats of the review: the overshoot gives a law an allowance equal to its own |a(0)|; […]
H27	In E4, the monotone relaxation of the model with the affine term is not a property of its first two moments: the Gaussian law that has the mean and the full covariance of the samples of W9 (G2_gaussianised_W9, fitted at each η on 8192 fresh samples of the […]	refuted	REFUTED on the second realisation of the data; TEST INVALID as written on the first (encadrant, 2026-11-13). Paired difference of the overshoot G2 - W9N: -0.0051 ± 0.0005 (main data) and -0.0056 ± 0.0005 (second realisation), 0 of 5 seeds positive: the three refutation clauses are met on both. Null control |overshoot(G3) - overshoot(G1)| < 0.02: […]
H28	In E4, a flow-matching model anchored on the Gaussian law with the full covariance (W14_full_anchor: v = v_G1(x, t, η) + network, the network being zero at initialisation so that the untrained model is G1) acquires a third-order part of the magnetic […]	registered	not reviewed yet
H29	In E4, recolouring the samples of W9 at sampling time onto the first and second moments fitted on the data (those of G1: W16_recoloured_full) gives a law with the second-order accuracy of G1 that keeps the third-order magnetic production of W9, and whose […]	registered	not reviewed yet
H30	In E4, at equal first and second moments, the non-Gaussian part of W9 lowers the passage of the magnetic energy above the reference at restart: […]	registered	not reviewed yet
H31	In E4, the excursion of the magnetic energy when the solver restarts from draws of G1 is not an effect of the interpolation of the covariance in η: […]	registered	not reviewed yet

Reading the two tables (Table 6.1 and its complement Table 5.1) by the place of the constraint:

- **By design.** Exact, and free at sampling time: the divergence is at round-off for every model on potentials. The design choice that changes the law is not the potential representation (W4 against W0) or the equivariance (W4s against W4, a gain of 46 floors out of 256) but the explicit linear part (W9, W11): a factor of twenty against the matched network without it.
- **By the loss.** The divergence penalty does not enforce its constraint and costs accuracy. The law-level stationarity loss lowers the residual of the energy identity, at parameter values without data, on a wrong model (0.29 to 0.19) and on a nearly right one (0.057 to 0.029); it does not lower the residual of the magnetic-helicity identity imposed with it; on the wrong model it leaves the distance to the reference unchanged, and on the nearly right one it increases it by three quarters.
- **By the sampling procedure.** Projections give exact zero divergence; projecting every stage brings nothing over projecting the final sample. The tilting collapses on a wrong

model and improves a nearly right one on most metrics. The per-sample rescaling satisfies the balance exactly, by removing the fluctuations of the energy budget that the true law has.

- **Model families.** At equal cost, flow matching interpolates in η better than the diffusion model, which is the better one at the training values. The two Gaussian laws are the baselines to beat. In energy distance they are not beaten: G1 is the closest law of the study and G0 the second. Among the laws close to the reference, G1 also has the lowest null-count error, while the learned models with the affine term (W9, W11) have a lower residual of the magnetic energy balance on the mean over the held-out η, through two errors that cancel at two values and add at the third, and are, among the laws close to the reference, those with a third-order part of the magnetic production (the anchored model W10 has one of 0.3 and 0.2 times that of the reference at the two lower η; the other variants were not measured; Section 5.5). The restart drifts of these tables are relative to each law's own initial energy and do not rank the laws: anchored on the reference energy, the six laws tested that start on it or below it gain magnetic energy after the restart, the Gaussian laws fitted to the data overshooting the reference by 9 and 17 % and the learned models with the affine term coming back to it from a 12 % deficit.

7 Toolbox

The offer asks for "reproducible code integrated into the project toolbox". The toolbox of this repository is the Python package `src/milady/`; what this work added to it is drawn in the code-backed diagram code-architecture.html (Figure 7.1) and listed in Table 7.1.

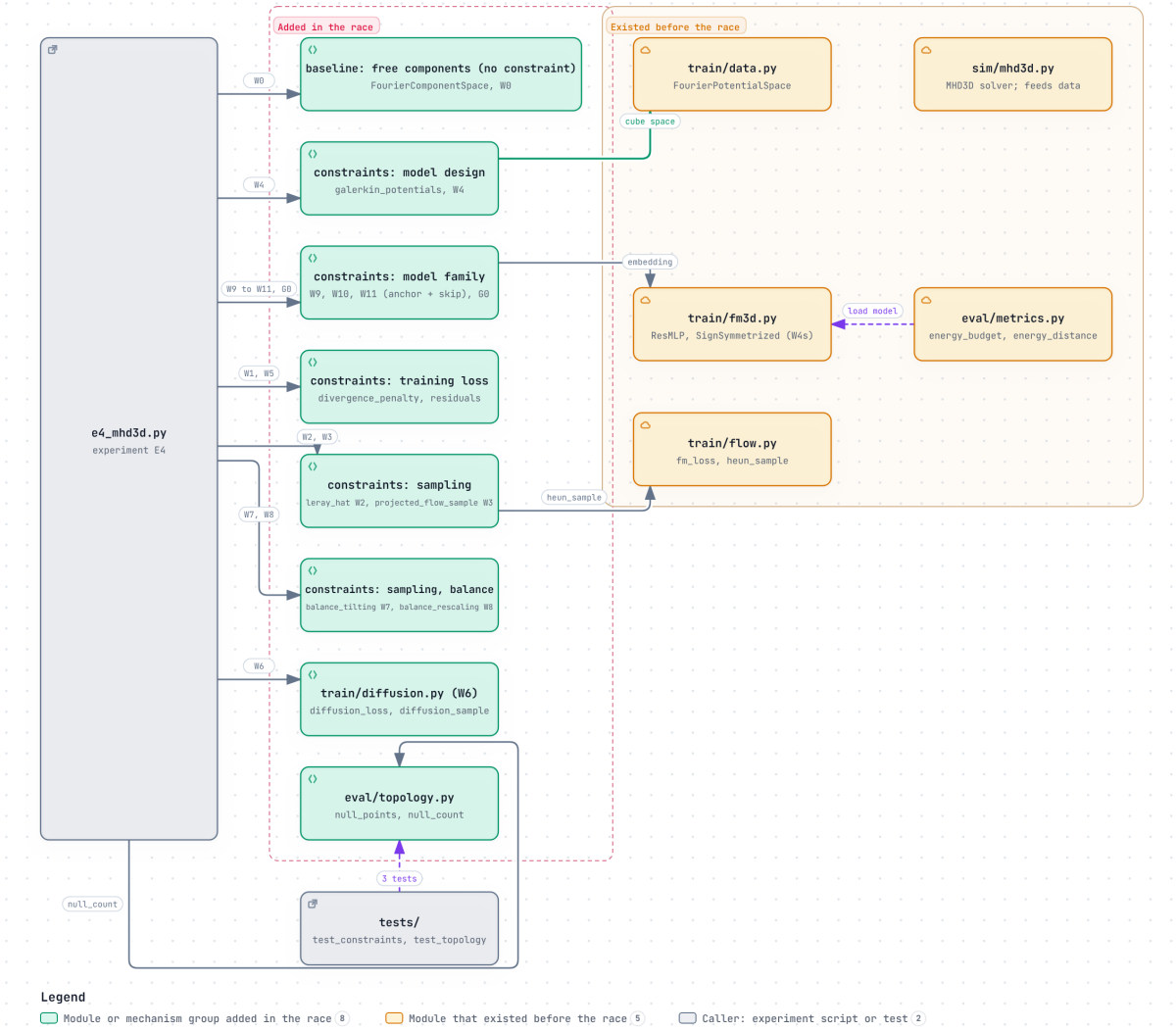


Figure 7.1: Code architecture of what E4 added to `src/milady/`: the constraint mechanisms by group, the diffusion module, the topology module, and the modules they build on. Static export of the Archify diagram `diagrams/code-architecture.html`.

Table 7.1: What entered the toolbox, with its tests.

Module	What it provides	Tests
`milady.train .constraints`	`REGISTRY` of 13 mechanisms: `FourierComponentSpace` (free components), `leray_hat`, `projected_flow_sample`, `divergence_penalty`, `budget_terms_hat` and `stationarity_residuals` (identities), `half_batch_product` (unbiased squared mean), `balance_tilting`, `balance_rescaling`, `ModewiseAffineSkip`, `ModewiseGaussian`, `FullGaussian`, `GaussianAnchoredField`	`tests/ test_constraints.py`: one test per mechanism (14 tests)
`milady.train .data`	`FourierPotentialSpace` with `truncation="cube"`: potentials on exactly the modes of the solver	`test_galerkin_potentials` in the same file
`milady.train .diffusion`	`diffusion_loss` (v-prediction, variance-preserving path), `diffusion_sample` (ancestral or deterministic)	`test_diffusion` in the same file
`milady.train .fm3d`	`ResMLP` and `ConditionEmbedding` with a `cond_scale` argument (largest frequency of the embedding of the condition)	existing tests
`milady.eval .topology`	`null_points`, `null_count`: null points of a periodic 3D field by the trilinear method	`tests/ test_topology.py` (3 tests, with the Poincaré–Hopf balance of signs)
`tools/ runpod_job.py`	`run`, `collect`, `exec`, `cleanup`, `spend`: GPU jobs with a time limit, a ledger and per-team pod names	`tests/ test_pilot_tools.py`

How to call it, for a latent space `space` and a trained field `model`:

```
from milady.train import constraints as C
from milady.train.data import FourierPotentialSpace
from milady.eval.topology import null_count

# latent space: 665 modes, 8 channels (vector potentials of u and B)
space = FourierPotentialSpace(n=16, kmax=5, truncation="cube")
fh = space.field_hat(space.denormalize(x))      # (batch, 2 fields, modes, 3)
# stationarity identities: energy, magnetic helicity, cross helicity
terms = C.budget_terms_hat(space, fh, nu=0.05, eta=0.05, forcing_hat=f_hat)
res = C.stationarity_residuals(terms, nu=0.05, eta=0.05)
# unbiased estimate of (E<j.B>)^2 over 4 conditions
loss = C.half_batch_product(*res["magnetic_helicity"], n_groups=4)
# Gaussian baselines and the explicit affine term of W9
gauss = C.ModewiseGaussian(sign=space.sign).fit(x_train, c_train)   # G0
full = C.FullGaussian(sign=space.sign).fit(x_train, c_train)        # G1
```

```
field = C.ModewiseAffineSkip(model, space.n_modes, space.n_channels, n_cond=1)
# null points of fields of shape (n, nx, ny, nz, 3)
counts = null_count(b_grid)
```

The tests check each mechanism against an independent computation: the identities against the time derivative of the solver, the Gaussian anchor against the closed form of Equation 3.4, the tilting against its moment conditions, the null finder against fields with a known number of nulls. `make test` runs them (the test suite, the linter and the check of the generated agent files pass on the branch).

The experiment itself is driven by scripts in `experiments/`: `e4_data.py` (data), `e4_mhd3d.py` (variants, by group), `e4_floor.py`, `e4_digest.py` and `digest.py` (digest), `race_metrics.py` (objectives), `e4_assemble_scan.py` (the width scan from its three jobs), `e4_plots.py` (the data plots of this report), `e4_report_tables.py` (its tables) and `export_explorer.py` (the data of the explorer).

8 Discussion

8.1 What each place of constraint buys

The three places named by the offer do not play the same role, and the measurements suggest a division of labour.

Design is the place for what is exactly known. A linear constraint that holds for every sample (zero divergence), an exact symmetry of the law ($B \to -B$), and the second-order structure of a homogeneous field (the mode-wise affine part of the velocity field) can all be built into the model. They cost nothing at sampling time and they cannot be violated. The first two did not change the quality of the law in this experiment; the third decided it.

The loss is the place for what is known about the law and not about each sample. The stationarity identities are of this kind. Imposed at the level of the law, with an unbiased estimator, the energy identity moved the residual it targets, at parameter values without data; the magnetic-helicity identity did not, and the term changed statistics it was not aimed at. The identities are necessary conditions, not a substitute for a model that fits the data: a model a hundred floors from the reference stays there, and a loss on one moment of a wrong law can only trade one error for another.

The sampling procedure is the place for linear per-sample constraints, and a dangerous place for the others. Projection onto a linear constraint is harmless. Enforcing a law-level identity on each sample is a different operation: it produced the best-looking energy balance of the comparison and a law without the fluctuations of the real one.

Three results go beyond the mechanisms.

The first is methodological. Two baselines without a network, cheap to compute, are closer to the reference in energy distance than every learned model of this report. Without the first, the comparison of constraints would have ranked models that were all wrong for the same reason. Without the second, which exists only because an external reviewer asked for it, the report would have credited the learned models with non-Gaussian physics on the strength of metrics that a Gaussian law improves just as much. A comparative study of generative models of a physical system needs its Gaussian baselines, including the one with the full covariance, before any claim about what a network has learned.

The second concerns evaluation. The energy distance on second-order features of single modes ranks a law without magnetic production (G0) above models that have it, and barely separates the best learned model from the Gaussian law with its first two moments (0.4 floors). The restart drift at one time, relative to the ensemble's own energy, confounds the bias of a law with its dynamics, and ranked laws that the anchored curves show to be comparably far from the reference energy over the course (0.13 and 0.10 for W9 and G1; 0.11 and 0.12 on the second realisation). The quantities that did discriminate are the split of the magnetic production into a mean-flow part and a third-order part (the residual of the balance alone hid two errors of W9 that cancel), the null count, and the restart curve anchored on the reference. The objectives of this pilot, written on the first day, rewarded the energy distance and a restart drift at one time; they would have been better with the production split and the anchored curve.

The third is about where the non-Gaussian part of the problem lies, and it is now partly **measured**. Three properties of the reference are out of reach of the Gaussian law with its own mean and covariance (G1): the third-order part of its magnetic production, a sink of more than a quarter of the Ohmic dissipation at the smallest η, which changes sign over the range; its number of null points, of which G1 has 2.1 times as many on the mean over η; and its stationarity under the solver, since the states of G1 at the held-out η, where it is interpolated, move 9 to 11 % away from the stationary magnetic energy. The best learned model holds about half of the first at two held-out values. On the second, its non-Gaussian part changes little: its null count differs from that of its own Gaussianisation by 0.04 out of 1.7. On the third, the overshoot registered for the restart does not separate the two, while the drift at $\tau = 4$ does (0.124 against 0.170). It is also behind G1 on the second-order part, by 10 to 16 % on the mean-flow production and 12 % on the magnetic energy. What a better generative model should learn is therefore, in this order, the covariance between modes to the accuracy of G1, and then the non-Gaussian part, for which the production split, the null count against the Gaussianised model and the anchored restart curve give three measurable targets.

8.2 Negative results and refuted claims

The register keeps what failed. Of the eleven hypotheses of the first protocol, written before any model was trained, six were refuted and one was dropped:

- *Potentials are not closer to the reference than free components* (H4 refuted). The constraint by design gives exact zero divergence at no cost, but the law is not better for it; the theory brief already says that with a perfect field the free model, the projections and the potentials give the same law, so any difference is an approximation effect, and here it has the opposite sign to the one expected.
- *The projected flow is not better than a final projection* (H6 refuted).
- *Equivariance does not lower the polarity asymmetry* (H9 refuted): the asymmetry was already at the sampling noise.
- *The helicity spectrum is not bipolar at $k = 1$ against $k \geq 2$* (H11 refuted): the shell $k = 2$ is negative.
- *The magnetic-helicity identity is not the more violated one* (H12 refuted).
- *The null count does not scale as k_λ^3* (H13 refuted on the slope).
- *Loss against tilting* (H8) was dropped: the criterion could be met by the sampling noise of one weighted ensemble.

The later hypotheses added five refutations. The embedding of the condition is not the cause of the failure of the first models (H15). Nor is the width of the network alone: a wider network is closer up to a point, and the widest one, with the lowest loss, is farther again (H21). Anchoring the velocity field on the mode-wise Gaussian field does not give a model at least as close as that law, with or without a learned affine correction (H19, H20). And the stationarity loss, applied to a model close to the reference, degrades the law beyond the margin that was registered (H23). Of the three hypotheses of the last tests, two were refuted, in opposite directions: the model with the affine term does hold a third-order part of the magnetic production (H25 predicted none), and, on the measure registered for the test (the largest deviation from the reference energy beyond the initial one), the Gaussian law with its first two moments does not move farther than the model does (H27 predicted that it would; refuted on one realisation of the data, the null control of the test failing on the other). That measure cannot see a passage above the reference smaller than the 12 % initial deficit; computed after the fact, the Gaussian law passes 2.4 % higher (2.6 % on the second realisation).

The interpretation of the main result was itself refuted. The first version of this report read the advantage of the flow-matching models over the mode-wise Gaussian law, on the magnetic energy balance, the null points and the restart test, as evidence that they had learned non-Gaussian, third-order structure. The `jury-externe` role showed with an oracle that a Gaussian law with the full covariance had the same advantage; the hypothesis was then registered, the baseline G1 built on the training values only and run, and the reading was replaced by the one of Section 5.4. The review that caught it is in `research/reviews/2026-11-13-jury-externe-report-v1.md`. The same role then showed that the restart drift of the objectives is confounded with the bias of each law on the magnetic energy (`research/reviews/2026-11-13-jury-externe-report-v3-and-plan.md`), and the statements of the second and third versions that ranked the laws by that drift were withdrawn (Section 5.5).

Two claims made during the work were corrected by the reviewing role and are reported as corrected. The first version of the objective on the energy distance said that a perfect generator would give a ratio of one; with a floor computed from trajectory halves it would give about 0.6, and the note was corrected before any model result existed. The statement "the cause of the failure is the width of the network" was refused as a conclusion from the first runs and registered as a hypothesis with a width scan (H21, Section 5.7).

A run labelled as a replication was not one. The job `e4_main` reproduced the first GPU run bit for bit, because the data and model seeds were the same and the hardware identical; it is a reproducibility check. The replication with other data seeds is the one of Section 5.9.

8.3 Limits

- **Scale.** The system has 665 Fourier modes and $Re = 10$. The flow is smooth, the statistics of the Fourier modes are close to Gaussian at second order, and that is why a mode-wise Gaussian law is such a strong baseline here. Nothing guarantees this at the Reynolds numbers of the MilaDy simulations, where the non-Gaussian part of the law carries the physics (intermittency, coherent structures). The ranking of the mechanisms may change with scale; the method of comparison does not.
- **Architecture.** The network acts on the flattened vector of modes. This does not scale: at 64^3 the latent has more than 10^5 dimensions. The explicit affine term is mode-wise and scales linearly with the number of modes, but the nonlinear part needs an architecture with locality in Fourier or physical space (a convolutional or attention-based field, as in the references of the section State of the art).
- **Homogeneity.** The mode-wise form of the affine term and of the Gaussian baseline G0 treats different Fourier modes as uncorrelated, which is true of a statistically homogeneous field. It is not true here: the forcing has a fixed phase, the mean flow is strong, and the covariance between modes carries the mean magnetic production (Section 3.2). In a bounded domain (the von Kármán geometry of the project) the departure from homogeneity is larger still, and the natural basis is different.
- **Metrics.** The energy distance is computed on 21 physical features, all second-order quantities of one snapshot. A law can be close to the reference in this distance and wrong in its higher-order structure: the distance barely separates the best learned model from its Gaussianisation (0.43 floors, about twice the sampling value; the Gaussianised law is 0.6 to 0.9 floors farther from the reference; Section 5.5). The production split sees the difference clearly, and the restart drift at $\tau = 4$ by 0.045; the null count and the overshoot registered for the restart hardly do. The reference null count itself is measured on 400 snapshots and

carries about 3 % of subsampling uncertainty that the standard errors over the seeds do not include (44.1 and 42.7 at the smallest held-out η in two runs on the same data).

- **One conditioning parameter.** Only η varies, over a range in which the dynamo is always saturated and the law changes smoothly. The dynamo threshold, where the law changes fastest (Proposition 9 of the theory brief), is outside the range.
- **Statistics.** Five seeds per variant; one realisation of the reference data for the main run and one independent realisation for the replication. The objective on the distance to the reference law is met inside the uncertainty of the noise floor.
- **Restart test.** The metric of the objectives measures the drift at one time, $\tau = 4$, relative to the ensemble's own initial energy; it is confounded with the bias of the law on the energy. The anchored curves to $\tau = 16$ exist for the Gaussian laws, the two models with the affine term and the anchored model W10 only; the anchored values quoted for two other variants come from older runs and an approximate initial energy, good to ± 0.01; for the other variants the tables give the single-time drift, which should not be read as a ranking. The rise of the magnetic energy at restart of the six laws tested that start on the reference or below it is measured and not explained; the overshoot registered for H26 and H27 gives a law an allowance equal to its initial deviation, and the null control of H27 failed on one of the two realisations of the data, which the review traced to the sampling noise of the initial energy on 128 restart samples.

8.4 What the incidents of the pilot teach

Four incidents cost time and are part of the result, because the race is also a test of the method. Rented GPUs were refused for lack of capacity for about an hour. The first pod ran for 24 minutes without its status being read, because a proxy refused the default user agent of the HTTP client. Another session sharing the cloud account deleted two running pods with its clean-up command. The container of the session restarted twice and orphaned running jobs. Each was repaired in the shared tool with a test (user agent, per-team pod names, the `collect` command), and announced in the journal. One bug was in the experiment script itself: a loop variable overwrote the suffix of the result files after a certain variant, and one run brought back incomplete files; it was found by reading what a job returned instead of trusting its exit code, fixed, and covered by a static test.

9 Conclusion

This report compared, on a small parametric MHD dynamo, the three places where a physical constraint can enter a generative model. All statements below are **measured** on this system: five seeds for the models, a jackknife over the 24 trajectories for the reference, and held-out parameter values unless said otherwise.

1. Zero divergence is obtained exactly by design (potentials) or by projection at sampling time; a penalty in the loss does not enforce it and degrades the law.
2. None of these divergence mechanisms decides the quality of the generated law. What decides it is whether the velocity field of the flow-matching model can carry the linear, second-order part of the transport: an explicit mode-wise affine term brings the same network from 205 noise floors to about 9.
3. In energy distance, the learned models do not beat two Gaussian baselines without a network. A Gaussian law per Fourier mode is at 3.5 floors but has no magnetic production, because it has no covariance between modes. A Gaussian law with that covariance, fitted on the training values and interpolated, is the closest law of the study (2.5 floors), with a magnetic energy balance residual of 0.07 and half the null-count error of the first. The learned models with the affine term have a lower residual of the magnetic energy balance on the mean (0.055 against 0.072), through two errors that cancel at the two lower values and add at the largest (item 5).
4. A law-level stationarity loss can be imposed at parameter values without data. It lowers the residual of the energy identity, on a wrong model and on a nearly right one, and it is not free: on the nearly right model it moves the law away from the reference. Exponential tilting on the same identities, at sampling time, collapses on a wrong model and improves a nearly right one on most metrics. Imposing a balance on each sample satisfies it exactly and removes real fluctuations.
5. The reference law is non-Gaussian in its magnetic production: the third-order part goes from $+0.14$ to -0.31 of the Ohmic dissipation over the range of η. The model with the affine term holds 0.6 and 0.4 of it at two held-out values and a small part of the opposite sign at the third; the Gaussian law fitted to its samples holds none, and the energy distance separates the two by 0.4 floors only.
6. No law close to the reference, Gaussian or learned, reproduces the magnetic topology (they have about twice the reference number of null points or more on the mean over η; the lowest ratio at a held-out η, G1 at the smallest, is 1.8, 1.7 on the second realisation, and 1.6 at the training $\eta = 0.03$), and none produces states that the solver accepts as stationary. Anchored on the reference energy, the six laws tested that start on it or below it gain magnetic energy after a restart: the Gaussian laws fitted to the data start on the reference and overshoot it by 9 and 17 % before coming back towards it, the learned model starts 12 % low and comes back to it. The restart drift of the objectives does not tell these apart. The Gaussian law with the moments of the reference has 2.1 times its null points on the mean over η: the topology of the reference is a non-Gaussian property, and so is its stationarity as far as that law at interpolated η shows.

What remains **conjectured**: that the degradation of the wide and long-trained networks is memorisation of the training snapshots (the width alone does not explain the failure of the plain

networks: H21 was refuted); that the third-order production held by the learned model may contribute to keeping it from passing above the reference energy at restart, as its Gaussianisation does by 2.4 % (computed after the fact; the order of the differences per η is the opposite of what this predicts); that a model with the covariance of G1 and the third-order production of the reference would be closer to stationary under the solver (which statistic of the magnetic field sets its number of null points is not known: the third-order production of W9 changes its count by 0.04 out of 1.7); and that the ranking of the mechanisms carries over to turbulent regimes.

The next steps for the internship follow from these results. Keep the Gaussian law with the full covariance as the baseline. Whether it can also anchor the flow-matching field, so that the network has only the non-Gaussian part to learn, is a hypothesis, registered as H28 and not yet run (as are the recolouring of the next sentence, H29, the restart at equal moments, H30, and the restart of the Gaussian law at a training η, H31): the two anchored models of this report did not stay as close as the law they were anchored on (H19, H20), and a full sample covariance cannot be estimated at 64^3. Correct the second moments of a learned model at sampling time, by recolouring its samples to the data-fitted covariance, and restart it at the reference energy. Use an architecture with locality for the correction, since the flattened perceptron does not scale. Put among the objectives the two parts of the production split, each against the reference, and the largest deviation from the reference energy over the anchored curve, initial deviation included, and report every model next to its Gaussianisation. Register the passage of the Gaussianised model above the reference as a hypothesis, signed, per η, with all 512 samples restarted. Find out why Gaussian states with the right covariance leave the stationary energy at restart, starting with the Gaussian law at a training η, without interpolation. Repeat the protocol on a 64^3 box at higher Reynolds numbers, where the law is less Gaussian and the solver of the toolbox runs at a few milliseconds per step on a GPU, and over a range of η that crosses the dynamo threshold. And look for training signals aimed at topology, since topology is where every model fails.

10 References

Albergo, M. S., and E. Vanden-Eijnden. 2023. "Building Normalizing Flows with Stochastic Interpolants." In *ICLR 2023*.

Alexakis, A. 2011. "Searching for the Fastest Dynamo: Laminar ABC Flows." *Physical Review E* 84: 026321. https://doi.org/10.1103/PhysRevE.84.026321.

Baldan, G., Q. Liu, A. Guardone, and N. Thuerey. 2026. "Physics Vs Distributions: Pareto Optimal Flow Matching with Physics Constraints."

Bastek, J.-H., W. Sun, and D. M. Kochmann. 2025. "Physics-Informed Diffusion Models." In *ICLR 2025*.

Brandenburg, A. 2001. "The Inverse Cascade and Nonlinear Alpha-Effect in Simulations of Isotropic Helical Hydromagnetic Turbulence." *Astrophysical Journal* 550: 824–40. https://doi.org/10.1086/319783.

Brandenburg, A., and K. Subramanian. 2005. "Astrophysical Magnetic Fields and Nonlinear Dynamo Theory." *Physics Reports* 417: 1–209. https://doi.org/10.1016/j.physrep.2005.06.005.

Cai, Z., Y. Cao, Y. Huang, and X. Zhou. 2026. "Weak Generative Sampler to Efficiently Sample Invariant Distribution of Stochastic Differential Equation." *SIAM Journal on Scientific Computing* 48 (4): C708–35. https://doi.org/10.1137/24M1665271.

Cheng, C., B. Han, D. C. Maddix, A. F. Ansari, A. Stuart, M. W. Mahoney, and Y. Wang. 2025. "Gradient-Free Generation for Hard-Constrained Systems." In *ICLR 2025*.

Christopher, J. K., S. Baek, and F. Fioretto. 2024. "Constrained Synthesis with Projected Diffusion Models." In *NeurIPS 2024*. https://doi.org/10.52202/079017-2834.

Chung, Kim, McCann, Klasky, and Ye. 2023. "Diffusion Posterior Sampling for General Noisy Inverse Problems." In *ICLR 2023*.

Csiszár, I. 1975. "I-Divergence Geometry of Probability Distributions and Minimization Problems." *Annals of Probability* 3 (1): 146–58. https://doi.org/10.1214/aop/1176996454.

Du, P., M. H. Parikh, X. Fan, X.-Y. Liu, and J.-X. Wang. 2024. "Conditional Neural Field Latent Diffusion Model for Generating Spatiotemporal Turbulence." *Nature Communications* 15: 10416. https://doi.org/10.1038/s41467-024-54712-1.

Galloway, D., and U. Frisch. 1986. "Dynamo Action in a Family of Flows with Chaotic Streamlines." *Geophysical and Astrophysical Fluid Dynamics* 36 (1): 53–83. https://doi.org/10.1080/03091928608208797.

Genuist, W., É. Savin, F. Gatti, and D. Clouteau. 2026. "Divergence-Free Diffusion Models for Incompressible Fluid Flows."

Haynes, A. L., and C. E. Parnell. 2007. "A Trilinear Method for Finding Null Points in a Three-Dimensional Vector Space." *Physics of Plasmas* 14 (8): 082107. https://doi.org/10.1063/1.2756751.

Ho, J., A. Jain, and P. Abbeel. 2020. "Denoising Diffusion Probabilistic Models." In *NeurIPS 2020*.

Huang, J., G. Yang, Z. Wang, and J. J. Park. 2024. "DiffusionPDE: Generative PDE-Solving Under Partial Observation." In *NeurIPS 2024*, 130291–323. https://doi.org/10.52202/079017-4140.

Kerrigan, Migliorini, and Smyth. 2024. "Functional Flow Matching." In *AISTATS 2024, PMLR 238*, 3934–42.

Köhler, J., L. Klein, and F. Noé. 2020. "Equivariant Flows: Exact Likelihood Generative Learning for Symmetric Densities." In *ICML 2020, PMLR 119*, 5361–70.

Li, Buzzicotti, Bonaccorso, and Biferale. 2026. "Physics-Constrained Diffusion Model for Synthesis of 3D Turbulent Data."

Lienen, Lüdke, Hansen-Palmus, and Günnemann. 2024. "From Zero to Turbulence: Generative Modeling for 3D Flow Simulation." In *ICLR 2024*.

Lim, J. H., N. B. Kovachki, R. Baptista, C. Beckham, K. Azizzadenesheli, J. Kossaifi, V. Voleti, et al. 2025. "Score-Based Diffusion Models in Function Space." *Journal of Machine Learning Research* 26 (158): 1–62.

Lipman, Chen, Ben-Hamu, Nickel, and Le. 2023. "Flow Matching for Generative Modeling." In *ICLR 2023*.

Liu, X., C. Gong, and Q. Liu. 2023. "Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow." In *ICLR 2023*.

Parnell, C. E., J. M. Smith, T. Neukirch, and E. R. Priest. 1996. "The Structure of Three-Dimensional Magnetic Neutral Points." *Physics of Plasmas* 3 (3): 759–70. https://doi.org/10.1063/1.871810.

Ponty, Y., P. D. Mininni, D. C. Montgomery, J.-F. Pinton, H. Politano, and A. Pouquet. 2005. "Numerical Study of Dynamo Action at Low Magnetic Prandtl Numbers." *Physical Review Letters* 94: 164502. https://doi.org/10.1103/PhysRevLett.94.164502.

Pouquet, A., U. Frisch, and J. Léorat. 1976. "Strong MHD Helical Turbulence and the Nonlinear Dynamo Effect." *Journal of Fluid Mechanics* 77 (2): 321–54. https://doi.org/10.1017/S0022112076002140.

Richter-Powell, Lipman, and Chen. 2022. "Neural Conservation Laws: A Divergence-Free Perspective." In *NeurIPS 2022*, 38075–88. https://doi.org/10.52202/068431-2759.

Salimans, T., and J. Ho. 2022. "Progressive Distillation for Fast Sampling of Diffusion Models." In *ICLR 2022*.

Shu, D., Z. Li, and A. Barati Farimani. 2023. "A Physics-Informed Diffusion Model for High-Fidelity Flow Field Reconstruction." *Journal of Computational Physics* 478: 111972. https://doi.org/10.1016/j.jcp.2023.111972.

Song, Sohl-Dickstein, Kingma, Kumar, Ermon, and Poole. 2021. "Score-Based Generative Modeling Through Stochastic Differential Equations." In *ICLR 2021*.

Székely, G. J., and M. L. Rizzo. 2013. "Energy Statistics: A Class of Statistics Based on Distances." *Journal of Statistical Planning and Inference* 143 (8): 1249–72. https://doi.org/10.1016/j.jspi.2013.03.018.

Utkarsh, Cai, Edelman, Gómez-Bombarelli, and Rackauckas. 2025. "Physics-Constrained Flow Matching: Sampling Generative Models with Hard Constraints." In *NeurIPS 2025*. https://doi.org/10.52202/085713-5354.

Wang, S., Z. Dou, S. Shan, T.-R. Liu, and L. Lu. 2026. "FunDiff: Diffusion Models over Function Spaces for Physics-Informed Generative Modeling." *Nature Communications* 17: 5749. https://doi.org/10.1038/s41467-026-72292-0.

Woltjer, L. 1958. "A Theorem on Force-Free Magnetic Fields." *Proceedings of the National Academy of Sciences* 44 (6): 489–91. https://doi.org/10.1073/pnas.44.6.489.

11 Reproducibility

11.1 Commands

Everything is in the branch `claude/modest-newton-zx8fqs` of the repository. From the root:

```
make test                                    # tests of src/milady and of the tools
cd experiments
python3 e4_data.py --quick                   # reference data (--jobs N: full data)
python3 e4_mhd3d.py --reference --quick      # reference statistics and floors
python3 e4_mhd3d.py --quick --seed 0 --group all   # groups A-E
python3 e4_mhd3d.py --quick --seed 0 --group D,G,M # G: Gaussian law G1; M: loss and
                                             # tilting on W9; L: long; S: width scan
python3 e4_floor.py                          # noise floors
python3 e4_assemble_scan.py                  # width scan from its three jobs
python3 digest.py > results/digest.md        # digest of E1-E4
python3 race_metrics.py --all                # objectives of the race
python3 e4_plots.py                          # data plots of this report
python3 e4_report_tables.py                  # tables of this report
python3 export_explorer.py --team claude     # data of the explorer
# on a GPU pod (5 seeds), from the root:
python3 tools/runpod_job.py run --hours 2 \
    --outputs experiments/results/replications/e4_final \
    -- bash experiments/e4_gpu_job.sh full "0 1 2 3 4" e4_final all
python3 tools/render.py teams/claude/report teams/claude/explorer
python3 tools/check_deliverables.py --team claude --built
```

Without `--quick`, the data take about 25 minutes on 11 CPU cores and one seed of the groups A to E about 35 minutes on a GPU. Quick outputs are not versioned and are never quoted.

11.2 Result files and commits

Table 11.1: Result sets, with the commit of the code that produced them and the commit that added them. The registrations precede the results of the hypotheses they concern.

Result set	Location	Code commit of the run	Commit of the results
Objectives of the team (day 1)	`race/teams/claude/objectives.toml`		`90e64af`

Result set	Location	Code commit of the run	Commit of the results
Register: H4–H14, H15, H16–H18, H19–H20, H21	`research/hypotheses.json`	–	`68e2fd4`, `9fd7272`, `fc04247`, `f3868f5`, `5027f7c`
Main run: groups A–E, 5 seeds (GPU job `e4_final`)	`experiments/results/e4_*.json`	`973fe3a`	`991d682`
First GPU run of groups A and B (reviewed for H4–H14)	`experiments/results/replications/e4_gpu`	`d0e018f`	`fb0786b`
Bitwise re-run of groups A and B	`experiments/results/replications/e4_main`	`62e31ca`	`8184bc9`
Convergence of the data	`experiments/results/replications/e4_convergence`	`d0e018f`	`56dbb91`
CPU seeds 0 and 1 (container)	`experiments/results/replications/e4_cpu`	`5ce4ad2`	`991d682`
Long trainings (H18): groups D, E, L	`experiments/results/replications/e4_final_long`	`973fe3a`	`bf07893`
Independent replication (other data seeds): groups A–E	`experiments/results/replications/e4_indep`	`d0e018f`	`d7aa2bd`
Register: H22, H23, H24	`research/hypotheses.json`	–	`a0cd918`
Gaussian law G1, loss and tilting on W9 (H22–H24): groups D, E, G, M	`experiments/results/replications/e4_gauss`	`6074dec`	`cc1da7b`
G1 on the second data realisation: groups D, G	`experiments/results/replications/e4_g1_indep`	`81a0e0c`	`133934a`
Width scan (H21), 5 seeds on three jobs: groups D, S	`experiments/results/replications/e4_scan2` (parts `e4_scan2a`, `b`, `c`)	`6074dec`, `d7aa2bd`, `830a40a`	`81a0e0c`, `133934a`
First width scan, cut by its time limit	`experiments/results/replications/e4_final_scan`	`973fe3a`	`d7aa2bd`
Register: H25, H26, H27	`research/hypotheses.json`	–	`bad65ae`
Production split, anchored restart, Gaussianised W9 (H25–H27): groups D, E, G, N, on both data realisations	`experiments/results/replications/e4_nongauss`, `e4_nongauss_indep`	`2e87dbc`	`245c783`

The main run was launched with a file-naming bug (the section Discussion): its result files were written under another suffix, copied back by a loop started on the pod, and renamed; the `encadrant` checked the renamed files against the per-seed logs of the job (`research/gpu-logs/e4-final/`).

11.3 GPU jobs and costs

Every GPU job of the pilot is in `research/compute-ledger.jsonl` (Table 11.2, Figure 11.1); the pods were rented on demand and deleted at the end of each job. Jobs without an exit code or with a non-zero one are the capacity failures, the interrupted jobs, the scan cut by its time limit and the incidents described in the section Discussion; the two negative entries are corrections made after another session deleted running pods, so that the ledger matches what was billed; one job (`e4_g1indep`) was launched twice by mistake.

Total: 30 ledger entries, 18.43 GPU hours, 17.08 USD.

Table 11.2: GPU jobs of the pilot (generated from `research/compute-ledger.jsonl`).

End (UTC)	GPU	USD/h	Hours	USD	Exit	Commit	Command (shortened)
11-12 17:48	RTX PRO 4500 Blackwell	0.72	0.404	0.29	–	`36908cc`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 17:58	RTX PRO 4500 Blackwell	0.72	0.026	0.02	–	`40f6188`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 17:59	RTX PRO 4500 Blackwell	0.72	0.026	0.02	–	`40f6188`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 18:18	RTX PRO 4500 Blackwell	0.72	0.104	0.07	–	`d4f2135`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 18:25	RTX PRO 4500 Blackwell	0.72	0.026	0.02	0	`f099fa8`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 18:27	RTX PRO 4500 Blackwell	0.72	0.026	0.02	0	`f099fa8`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 18:40	RTX A6000	0.69	0.026	0.02	0	`47d057d`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 18:44	RTX PRO 4500 Blackwell	0.72	0.146	0.11	0	`8fd79a0`	e4_gpu_job.sh quick 0
11-12 19:03	RTX PRO 4500 Blackwell	0.72	0.172	0.12	0	`d0e018f`	e4_convergence_job.sh
11-12 20:16	RTX PRO 4500 Blackwell	0.72	1.415	1.02	0	`d0e018f`	e4_gpu_job.sh full “0 1 2 3 4”
11-12 22:21	RTX PRO 4500 Blackwell	0.72	0.590	0.42	–	`1c21afc`	e4_gpu_job.sh full “0 1 2 3 4” e4_gpu_de D,E
11-12 22:23	RTX PRO 4500 Blackwell	0.72	1.820	1.31	–	`a20bcb1`	e4_gpu_job.sh full “0 1 2 3 4” e4_gpu_all
11-12 22:24	RTX PRO 4500 Blackwell	0.72	0.031	0.02	–	`837c23a`	e4_gpu_job.sh full “0 1 2 3 4” e4_main all
11-12 22:24	RTX PRO 4500 Blackwell	0.72	-0.590	-0.42	–	`1c21afc`	CORRECTION of the entry of pod hd6qx71mm6jekm: the pod was d …
11-12 22:24	RTX PRO 4500 Blackwell	0.72	-0.628	-0.45	–	`a20bcb1`	CORRECTION of the entry of pod 00t6s3l481flvm: the pod was d …
11-12 23:24	A100-SXM4-80GB	2.19	0.126	0.28	–	`5027f7c`	e4_gpu_job.sh full “0 1 2 3 4” e4_scan S
11-12 23:24	RTX PRO 4500 Blackwell	0.72	0.126	0.09	–	`5027f7c`	e4_gpu_job.sh full “0 1 2 3 4” e4_long L
11-12 23:27	RTX PRO 4500 Blackwell	0.72	0.983	0.71	0	`62e31c9`	collected after a restart of the session: milady-claude-job- …
11-13 00:17	RTX PRO 4500 Blackwell	0.72	0.879	0.63	0	`973fe3a`	collected after a restart of the session: milady-claude-job- …
11-13 00:55	RTX PRO 4500 Blackwell	0.72	1.508	1.09	0	`973fe3a`	collected after a restart of the session: milady-claude-job- …
11-13 01:25	GeForce RTX 4090	0.89	1.356	1.21	0	`dfc02a2`	env E4_DATA_SEED=100 e4_gpu_job.sh full “0 1 2 3 4” e4_indep …

End (UTC)	GPU	USD/h	Hours	USD	Exit	Commit	Command (shortened)
11-13 01:27	A100-SXM4-80GB	2.19	2.034	4.46	124	`973fe3a`	collected after a restart of the session: milady-claude-job- …
11-13 02:09	RTX PRO 4500 Blackwell	0.72	0.762	0.55	0	`607ccbd`	e4_gpu_job.sh full “0 1 2 3 4” e4_mech D,E,G,M
11-13 03:01	RTX PRO 4500 Blackwell	0.72	1.636	1.18	0	`607ccbd`	e4_gpu_job.sh full “0 1” e4_scan2a D,S
11-13 03:16	RTX PRO 4500 Blackwell	0.72	1.472	1.06	0	`d7aa2bd`	e4_gpu_job.sh full “2 3” e4_scan2b D,S
11-13 03:35	RTX PRO 4500 Blackwell	0.72	0.505	0.36	0	`81a0e0c`	env E4_DATA_SEED=100 e4_gpu_job.sh full “0 1 2 3 4” e4_g1ind …
11-13 03:38	RTX PRO 4500 Blackwell	0.72	0.548	0.39	0	`81a0e0c`	env E4_DATA_SEED=100 e4_gpu_job.sh full “0 1 2 3 4” e4_g1ind …
11-13 03:38	L40S	1.09	1.096	1.19	0	`830a40a`	collected after a restart of the session: milady-claude-job- …
11-13 06:19	RTX PRO 4500 Blackwell	0.72	0.882	0.64	0	`2e87cbc`	e4_gpu_job.sh full “0 1 2 3 4” e4_nongauss D,E,G,N
11-13 06:21	RTX PRO 4500 Blackwell	0.72	0.918	0.66	0	`2e87cbc`	env E4_DATA_SEED=100 e4_gpu_job.sh full “0 1 2 3 4” e4_nonga …

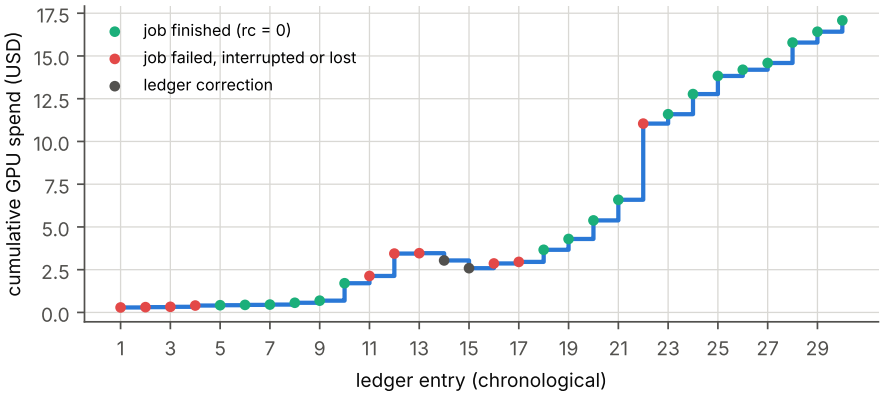


Figure 11.1: Cumulative GPU spend of the pilot, one point per entry of the compute ledger. Data plot from `research/compute-ledger.jsonl`.

The cost of the session itself (tokens, sub-agent calls, CPU time) is reported by `tools/usage_report.py` in `research/usage/latest.md`.

11.4 Figures of this report

The eight schematics were drawn with the `schematic-figure` skill (sources in `site/teams/claude/figures/<n` rendered to SVG and PDF), the three diagrams with Archify (`site/teams/claude/diagrams/`), and the data plots by `experiments/e4_plots.py` from the result files named in each caption (`site/teams/claude/plots/plots.json` lists them).